Appendix A — UML diagram set

Campus Shuttle Tracking & Demand Prediction · BCS 305 Software Architecture · Fall 2026

All diagrams are drawn with **PlantUML** (a text-based UML tool) using standard UML 2 notation. The editable sources are in `docs/uml/*.puml`; rendered SVG/PNG files go to `docs/uml/out/` (`npm run uml`). The class, component and sequence diagrams were written from the actual source code and were compared with it member by member on 1 November 2026 (corrections are listed in `docs/test-plan.md` §4). They were compared again in the owner review after PRs #12 and #13, which corrected diagrams 02, 04, 05, 07–12, 14 and 16 (`docs/test-plan.md` §4, item 13). **Rendering (2 November 2026):** all 16 sources render with the checksum-verified PlantUML MIT 1.2026.9 (built-in Smetana layout, no Graphviz). `npm run uml` validates every export (complete, unclipped, rendered from the current source) and records it in `docs/uml/out/manifest.json`; `npm run uml:check` repeats that check without Java. Every SVG is checked as well-formed XML and every PNG is fully decoded. In this PDF a diagram is never printed below 60 % of its rendered size: one that is too large for a page at that scale gets a vector overview page (zoom in on screen) followed by overlapping detail pages. On the reading pages most labels print at 6.2–9.8 pt, and the smallest annotations at 5.5–6.2 pt. These sizes were measured on screen; nobody has yet checked a printed copy (issue #1). Only one aspect is checked automatically: `test/architecture.test.ts` fails if a source file imports a component that the component diagram does not allow. It does **not** check UML operations, attributes, associations, multiplicities or sequence messages; those were reviewed by hand and can drift when the code changes. Where a diagram writes a type as `{…}`, the code uses an inline TypeScript object type with no class name.

#	Diagram	Required by the brief	Source file
A.1	Use case diagram	Yes	`01-use-case-diagram.puml`
A.2	Use case descriptions (5 primary use cases)	Yes (≥ 3)	this document
A.3	Activity diagram — Run a trip	Yes	`02-activity-run-trip.puml`
A.4	System sequence diagram — Run a trip and track it	Yes	`03-system-sequence-diagram.puml`
A.5	Design sequence diagram — processing a location upload	Extra	`04-sequence-location-update.puml`
A.6	State machine — Trip and GPS signal	"Where relevant"	`05-state-machine-trip.puml`
A.7	State machine — Waiting request	"Where relevant"	`06-state-machine-waiting-request.puml`
A.8	Domain model (conceptual class diagram)	Yes	`07-domain-model.puml`
A.9	Design class diagram 1/2 — tracking, network, real-time	Yes	`08-class-diagram-tracking.puml`
A.10	Design class diagram 2/2 — demand, forecasting, identity	Yes	`09-class-diagram-demand-identity.puml`
A.11	Component diagram	Yes	`10-component-diagram.puml`
A.12	Deployment diagram	Yes	`11-deployment-diagram.puml`
A.13–A.17	Design pattern class diagrams (Strategy, Observer, State, Repository, Adapter)	Yes (≥ 3 patterns)	`12`–`16-pattern-*.puml`

A.1 Use case diagram

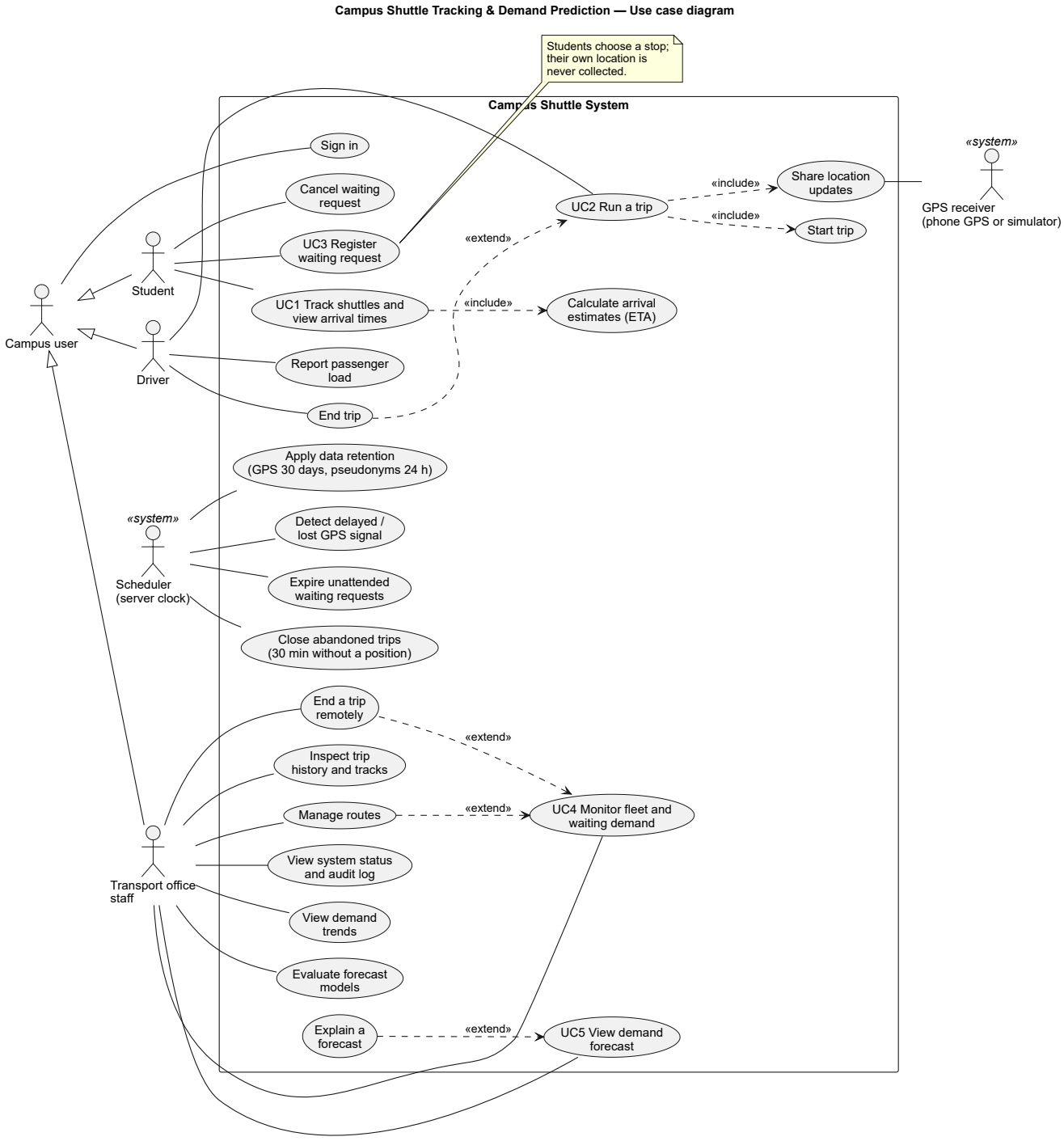


Figure A.1 — Use case diagram — overview at 55 % (read the 2 detail pages that follow, at 83 %; on screen, zoom in)

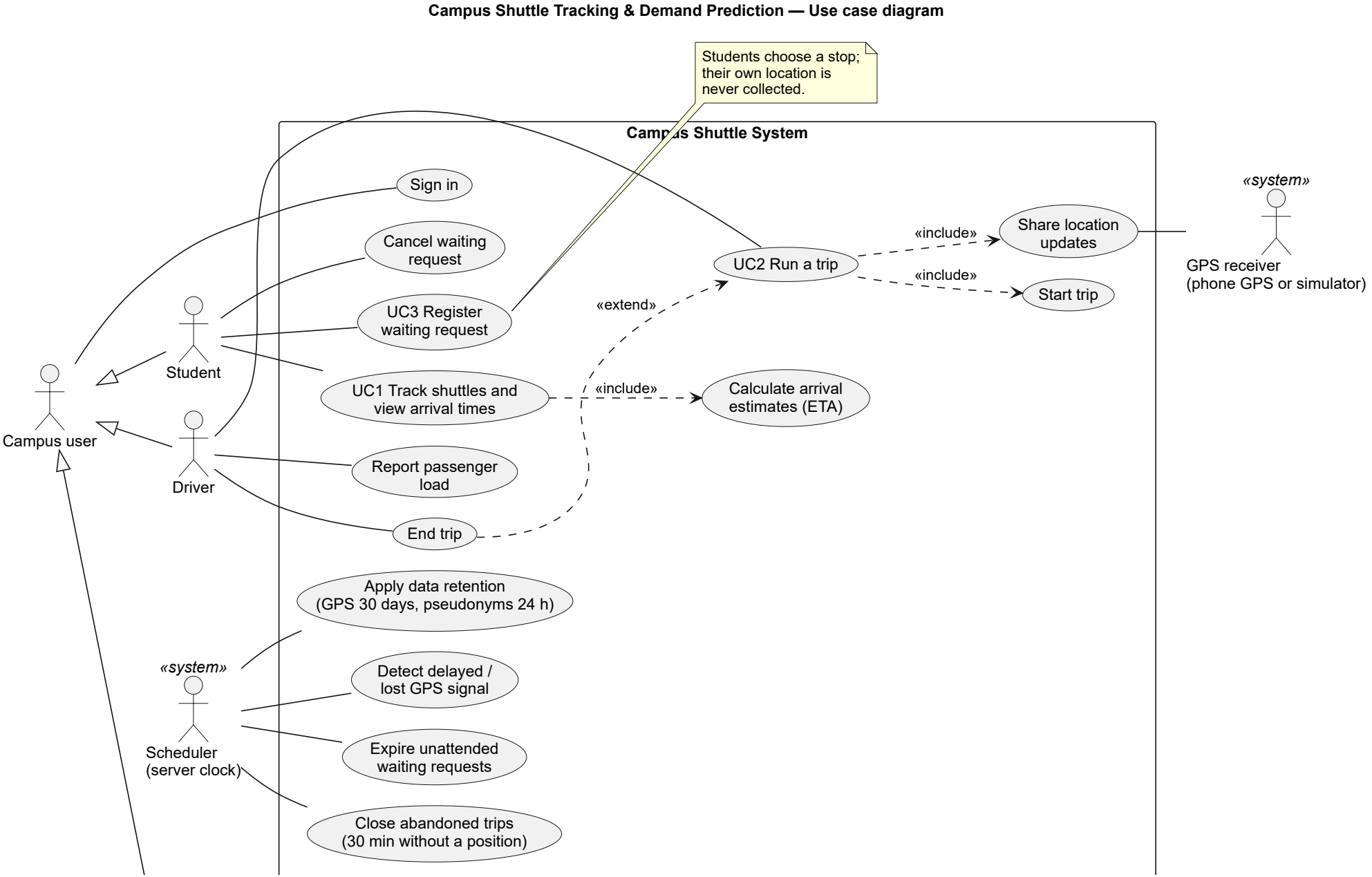


Figure A.1 — Use case diagram — detail 1 of 2 (row 1 of 2), at 83 %

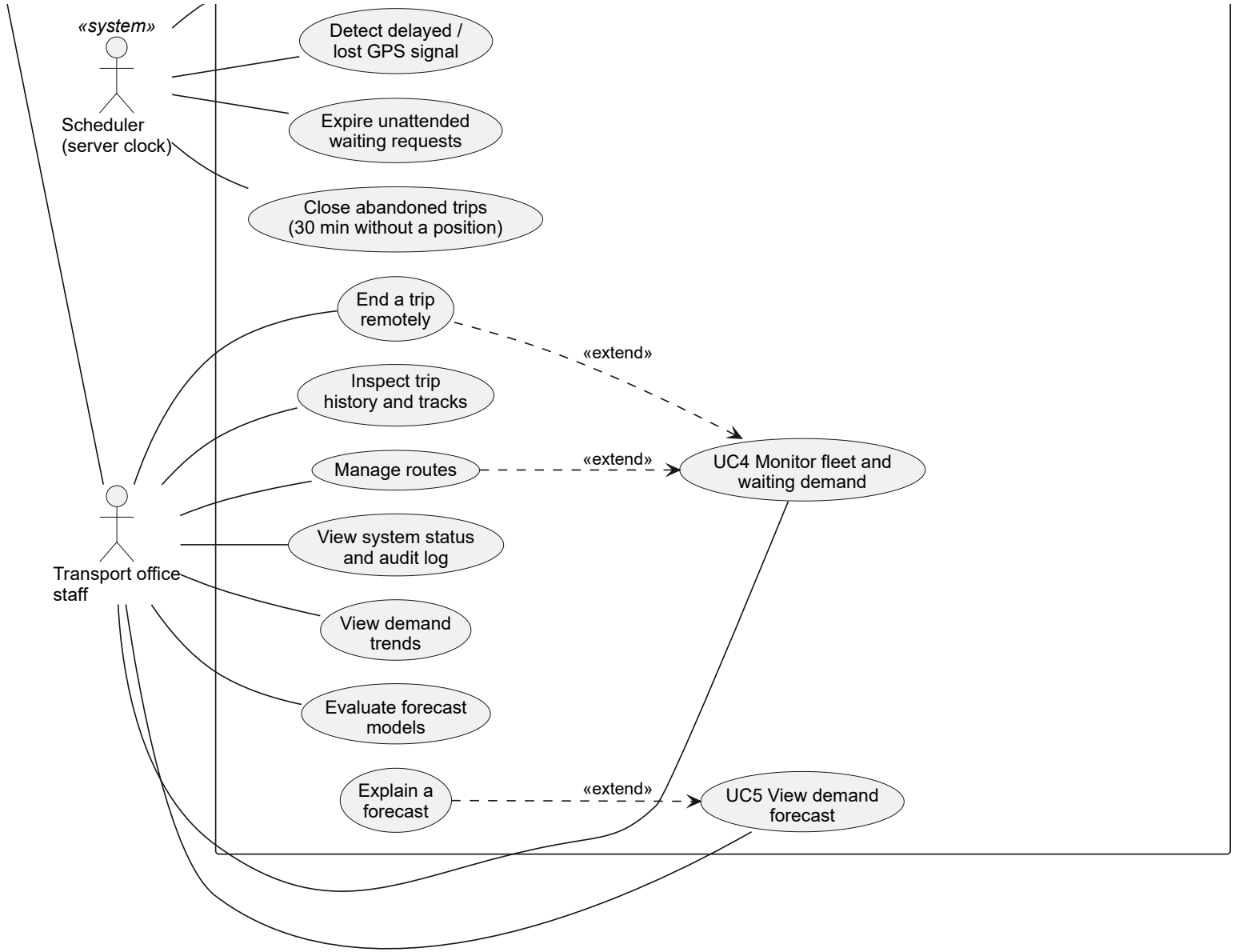


Figure A.1 — Use case diagram — detail 2 of 2 (row 2 of 2), at 83 %

Three primary human actors (Student, Driver, Transport office staff) share a generalised *Campus user* actor for sign-in. Two system actors appear as secondary actors: the **GPS receiver** (the phone's GPS or the built-in simulator) and the **Scheduler** (the server clock that detects stale GPS and expires waiting requests). `«include»` marks behaviour that always happens as part of a use case; `«extend»` marks optional behaviour.

A.2 Use case descriptions

The five primary use cases below are all implemented end to end in the prototype. UC1 and UC2 together form the main demonstration scenario.

UC1 — Track shuttles and view arrival times

Field	Description
Primary actor	Student
Stakeholders and interests	Student: wants to know whether to walk to the stop now, without waiting in the heat. Transport office: fewer phone calls asking "where is the bus?". Driver: wants passengers ready at the stop.
Preconditions	Student is signed in. At least one route is in service.
Trigger	Student opens the student page.
Main success scenario	1. System shows the schematic map with routes, stops and every active shuttle, and opens a live feed. 2. Student selects a stop (on the map or from a list; the choice is remembered on that device). 3. System lists every shuttle on a route serving that stop, soonest first, with the estimated minutes, clock time, distance, passenger load and a live-signal badge. 4. As drivers send positions, the system pushes updates; positions and minutes change without reloading. 5. When a shuttle on a loop route reaches the selected stop, its next-lap estimate is recalculated and the list is sorted again. A non-loop route does not invent a next lap.
Extensions	3a. No shuttle is running on a route serving the stop: system says so plainly. 4a. A shuttle's newest position is more than 20 s old: its estimate is shown as approximate ("~6 min") with "location delayed"; when the estimate time has passed it shows "Due". 4b. No position for more than 60 s: the system shows "GPS signal lost · last seen N min ago" and **no estimate**. 4c. The student's connection drops: a banner says "reconnecting — times may be out of date"; on reconnection a full snapshot replaces the old state. 4d. Session expired: banner offers to sign in again.
Postconditions	None (read-only use case).
Quality attributes	Freshness (QA-1), scalability with many viewers (QA-2), honesty under GPS loss (QA-3), privacy: no driver identity is ever sent to students (QA-6).
Implemented in	`public/js/student.js`, `src/web/routes/shared-routes.ts`, `src/realtime/sse-hub.ts`, `src/tracking/tracking-service.ts`

UC2 — Run a trip (start, share location, end)

Field	Description
Primary actor	Driver. Secondary actor: GPS receiver (device GPS or simulator).
Stakeholders and interests	Driver: minimal interaction while working. Students: accurate positions. Office: knows which vehicle is on which route and who is accountable.
Preconditions	Driver is signed in, has no other active trip, and the chosen shuttle is free.
Trigger	Driver taps "Start trip" after choosing route, shuttle and location source.

Field	Description
Main success scenario	1. System validates the route is in service and that the shuttle and driver are free, then creates an ACTIVE trip (signal "awaiting first GPS position") and announces it to all viewers. 2. Driver app starts the GPS source. 3. The simulator obtains a position every ~3 s (device GPS is event-driven and throttled to at most one fix per 2 s) and uploads all queued positions in one batch with the device's current time. 4. System corrects device clock skew, validates and orders the positions, stores them, snaps the shuttle onto the route, updates its smoothed speed, recalculates ETAs, detects stops reached, and pushes one update to all viewers. 5. App shows "what passengers see" (Live / Delayed / GPS lost) and the next stop. 6. Driver optionally sets passenger load (seats available / filling up / full). 7. Driver taps "End trip" and confirms; the app first delivers the queued positions (waiting at most 5 s), holds new ones and sends the end request; the system completes the trip and removes the shuttle from every map; only after the server confirms does the app stop GPS and show "Trip ended".
Extensions	1a. Shuttle or driver already on an active trip, or route out of service: system refuses with the reason (HTTP 409; an unknown route or shuttle gives 404). 3a. GPS produces no positions (tunnel, outage): nothing is uploaded; students see the delayed → lost sequence (UC1 4a–4b). 3b. Network unavailable: positions stay queued in the open driver page (newest 500 kept; older ones dropped and counted; lost if the page is closed or reloaded) and are retried with exponential backoff (1 s → 15 s); when the network returns they are delivered in batches of up to 50 and the shuttle jumps to its true position; positions more than 10 minutes old on arrival are rejected. 4a. A position is duplicated, out of order, physically impossible (> 110 km/h jump), too inaccurate (> 150 m), too old (> 10 min) or outside the service area: it is rejected with a reason code and the rest of the batch is still accepted. 4b. The page is reloaded: the trip resumes from the server's last known position with the same location source as its latest accepted position (device GPS or simulator). If no position has been accepted yet, the app shares nothing until the driver chooses a source. 7a. The office ended the trip remotely (UC4): the next upload returns 409/404 and the app returns to the start screen with an explanation. 7b. No accepted position for 30 minutes (counted from the last position, or from the trip start if there was none): the system closes the trip automatically (TERMINATED, reason NO_SIGNAL_TIMEOUT). 7c. The end request fails (no network, expired session, server unreachable): the trip stays ACTIVE, location sharing continues, the app says the trip was **not** ended and the driver can try again; if the server answers that the trip had already ended (409/404), the app treats it as ended.
Postconditions	The trip is COMPLETED or TERMINATED, with its GPS trace stored (deleted after 30 days).
Implemented in	`public/js/driver.js`, `public/js/gps.js`, `src/web/routes/driver-routes.ts`, `src/tracking/*`

UC3 — Register a waiting request ("I'm waiting here")

Field	Description
Primary actor	Student
Stakeholders and interests	Student: wants the shuttle to know someone is waiting. Office: real-time demand signal and history for forecasts. Privacy officer: no tracking of individual students.
Preconditions	Student is signed in and has selected a stop.
Main success scenario	1. Student taps "I'm waiting here", optionally choosing a specific route. 2. System records the request against a **pseudonymous reference** (keyed hash of the user id), sets a 20-minute expiry, and pushes the stop's new waiting count to all viewers. 3. The student sees "You're on the waiting list at Library · expires in 20 min". 4. When a shuttle on a matching route reaches the stop, the system marks the request SERVED and the student sees "A shuttle reached Library at 08:14".

Field	Description
Extensions	1a. Same stop and route requested again: the existing request is returned (idempotent). 1b. Different stop or route: the previous request is cancelled and replaced. 1c. Route does not serve the stop: rejected with a reason. 1d. More than 6 requests in an hour: rejected (429) to prevent abuse that would distort demand data. 3a. Student cancels: request CANCELLED and the count updates. The open page reconciles its private request after a matching waiting-count event and after a reconnect snapshot, including closures missed while offline. 4a. No shuttle within 20 minutes: request EXPIRED.
Postconditions	Request closed in one of SERVED / CANCELLED / EXPIRED; 24 h later its pseudonymous link is erased and it remains only as anonymous demand data.
Implemented in	`src/demand/waiting-service.ts`, `src/web/routes/student-routes.ts`, `public/js/student.js`

UC4 — Monitor the fleet and waiting demand (with remote trip termination)

Field	Description
Primary actor	Transport office staff
Preconditions	Staff member signed in with the office role.
Main success scenario	1. System shows KPI tiles (shuttles on trip, GPS problems, students waiting, live viewers), the fleet map with waiting badges, waiting counts per stop, and a table of active trips including the accountable driver, GPS state, last position age, load and next-stop ETA — all updated live. 2. Staff opens a trip's track to inspect its GPS trace: positions, distance, duration, longest gap and number of gaps longer than the stale threshold.
Extensions	1a. *End a trip remotely* («extend»): staff chooses "End", must give a reason; the system terminates the trip, records actor, time and reason in the audit log, and the driver app is told on its next contact. 1b. *Manage routes*: take a route out of service or back, edit name/colour/planned speed/dwell time, or create a new route from an ordered list of stops (created inactive for review). Every change is audited.
Implemented in	`public/js/office.js`, `src/web/routes/office-routes.ts`, `src/network/network-service.ts`

UC5 — View the demand forecast (with explanation)

Field	Description
Primary actor	Transport office staff
Main success scenario	1. Staff opens "Demand & forecast", picks a date and a model. 2. System shows the expected requests per hour (all stops), a stop × hour heat map, the peak hour and busiest stop, and states which data sources were used (synthetic and/or live). 3. Staff selects a cell; the system explains it: method, the exact past observations used (dates, counts, weights) and the plausible range. 4. Staff compares models in the backtest table (MAE, RMSE, bias, range coverage).
Extensions	2a. Not enough history for a stop-hour: the cell shows "–" and **no number is invented**. 2b. History contains synthetic data: a warning banner states that the figures illustrate the method only.
Implemented in	`src/demand/demand-service.ts`, `src/demand/prediction/*`, `public/js/office.js`

A.3 Activity diagram — Run a trip

Swimlanes separate the driver app from the server. The two interruption paths are visible: *GPS outage* (no position is appended to the queue) and *network loss* (the queue is kept and retried with backoff). Two further paths are shown: the server ending the trip while it runs (office or 30-minute auto-close), and ending the trip, where the outer loop repeats until the server confirms the end, so a failed request leaves the trip active.

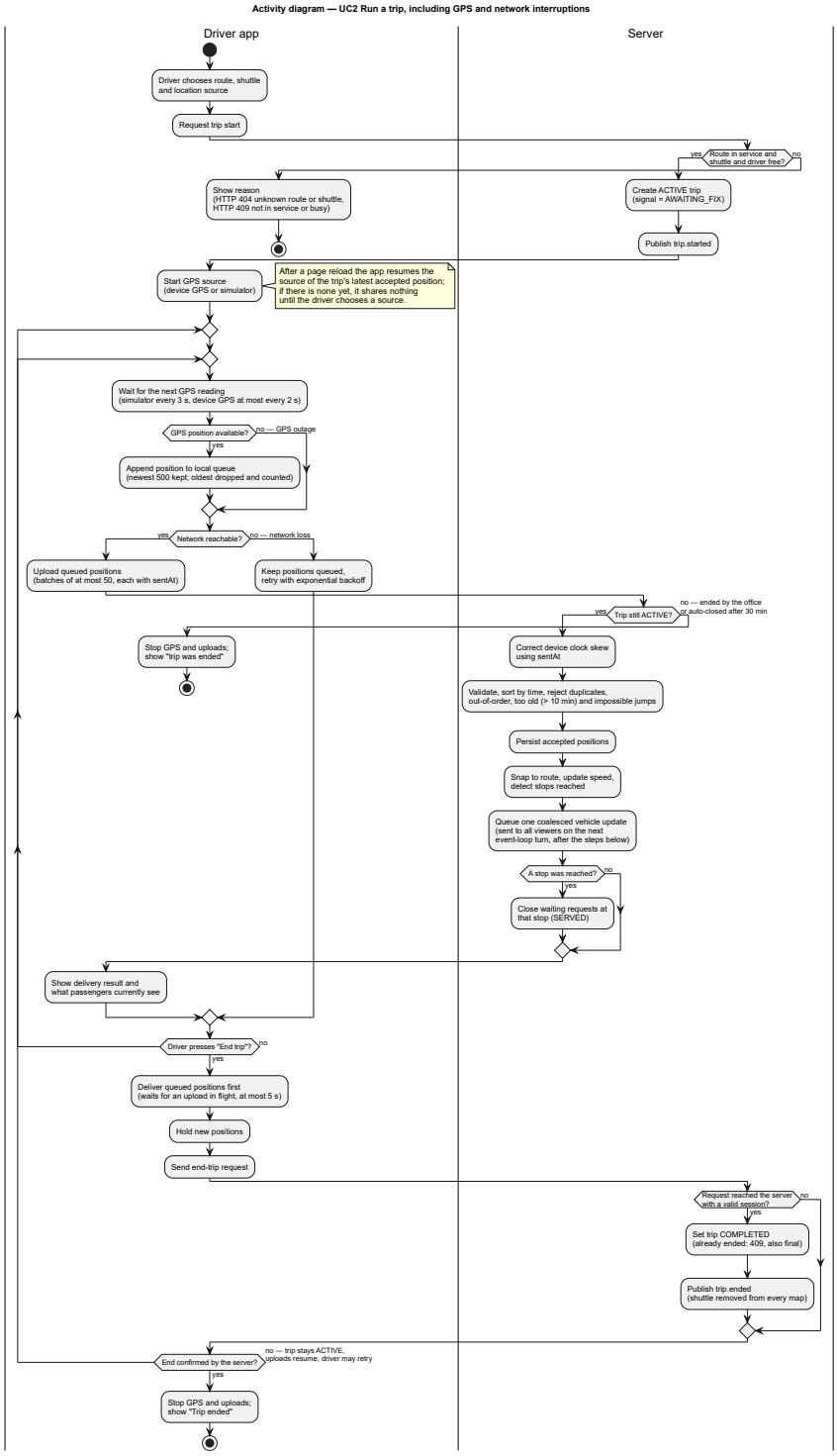


Figure A.3 — Activity diagram: Run a trip with GPS and network interruptions — overview at 34 % (read the 3 detail pages that follow, at 75 %; on screen, zoom in)

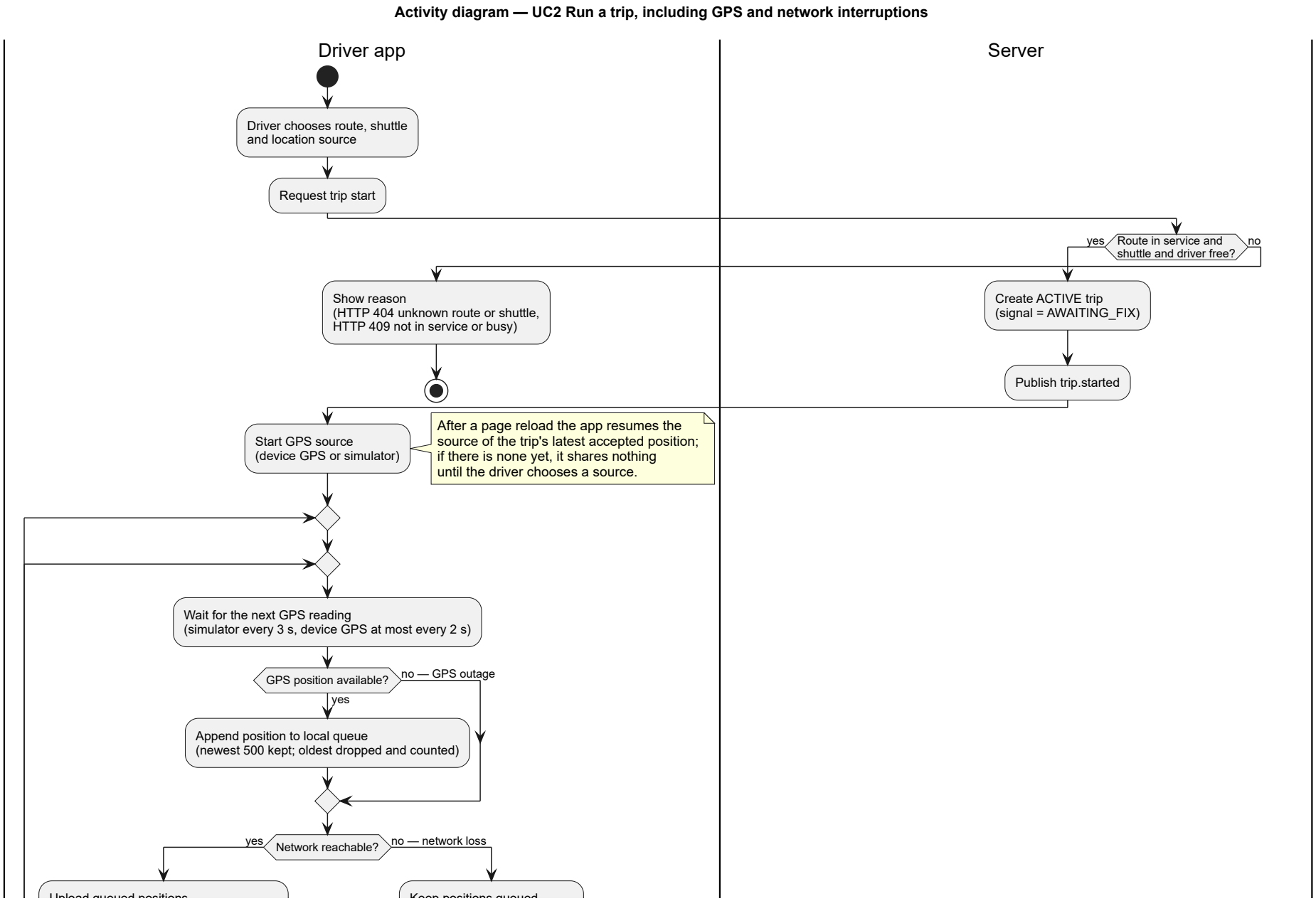


Figure A.3 — Activity diagram: Run a trip with GPS and network interruptions — detail 1 of 3 (row 1 of 3), at 75 %

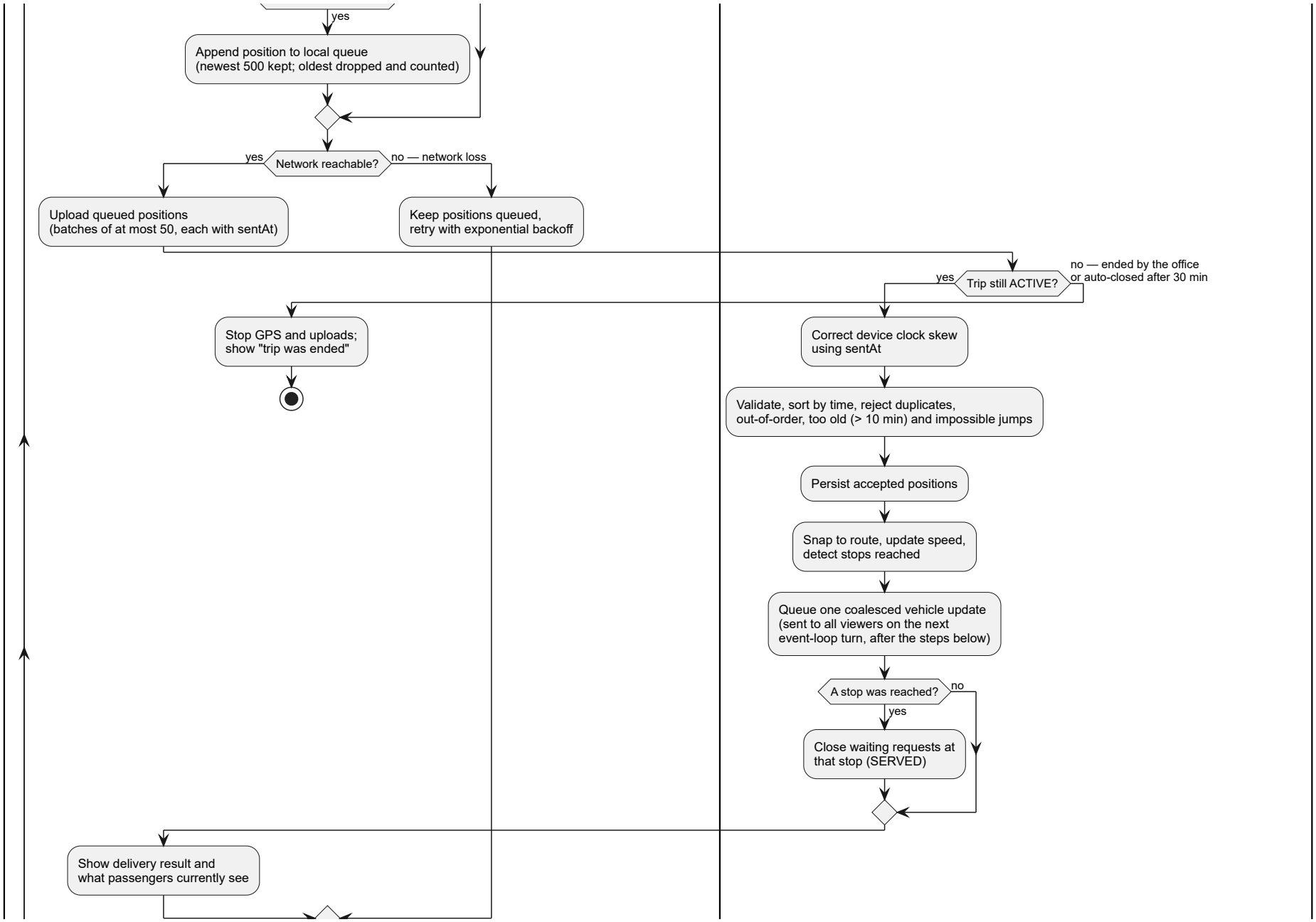


Figure A.3 — Activity diagram: Run a trip with GPS and network interruptions — detail 2 of 3 (row 2 of 3), at 75 %

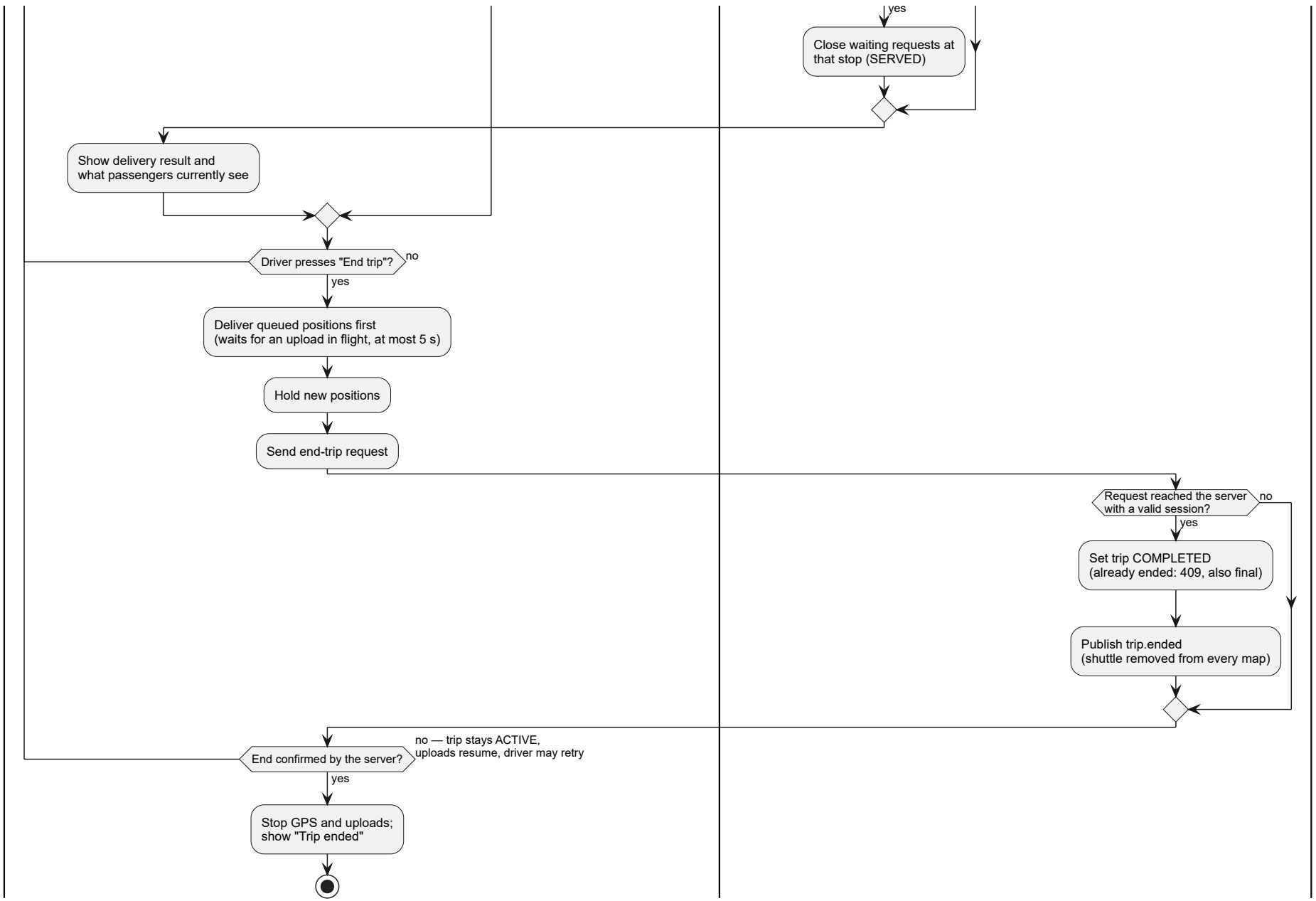


Figure A.3 — Activity diagram: Run a trip with GPS and network interruptions — detail 3 of 3 (row 3 of 3), at 75 %

A.4 System sequence diagram — Run a trip and track it

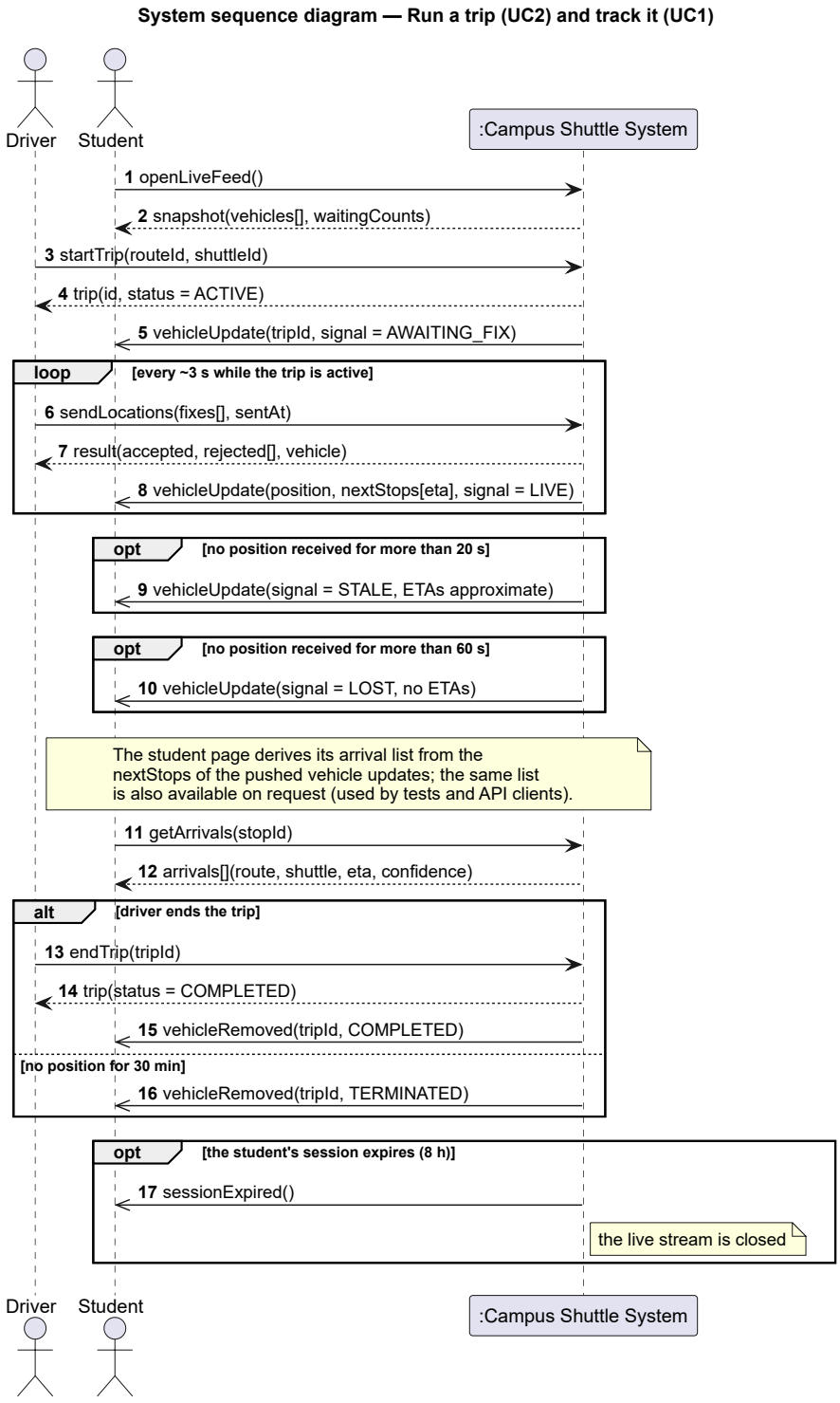


Figure A.4 — System sequence diagram for UC2 + UC1

The system is a black box; messages are the system operations of the REST API and the live feed. The `opt` fragments are system-initiated (time) events.

A.5 Design sequence diagram — processing one location upload

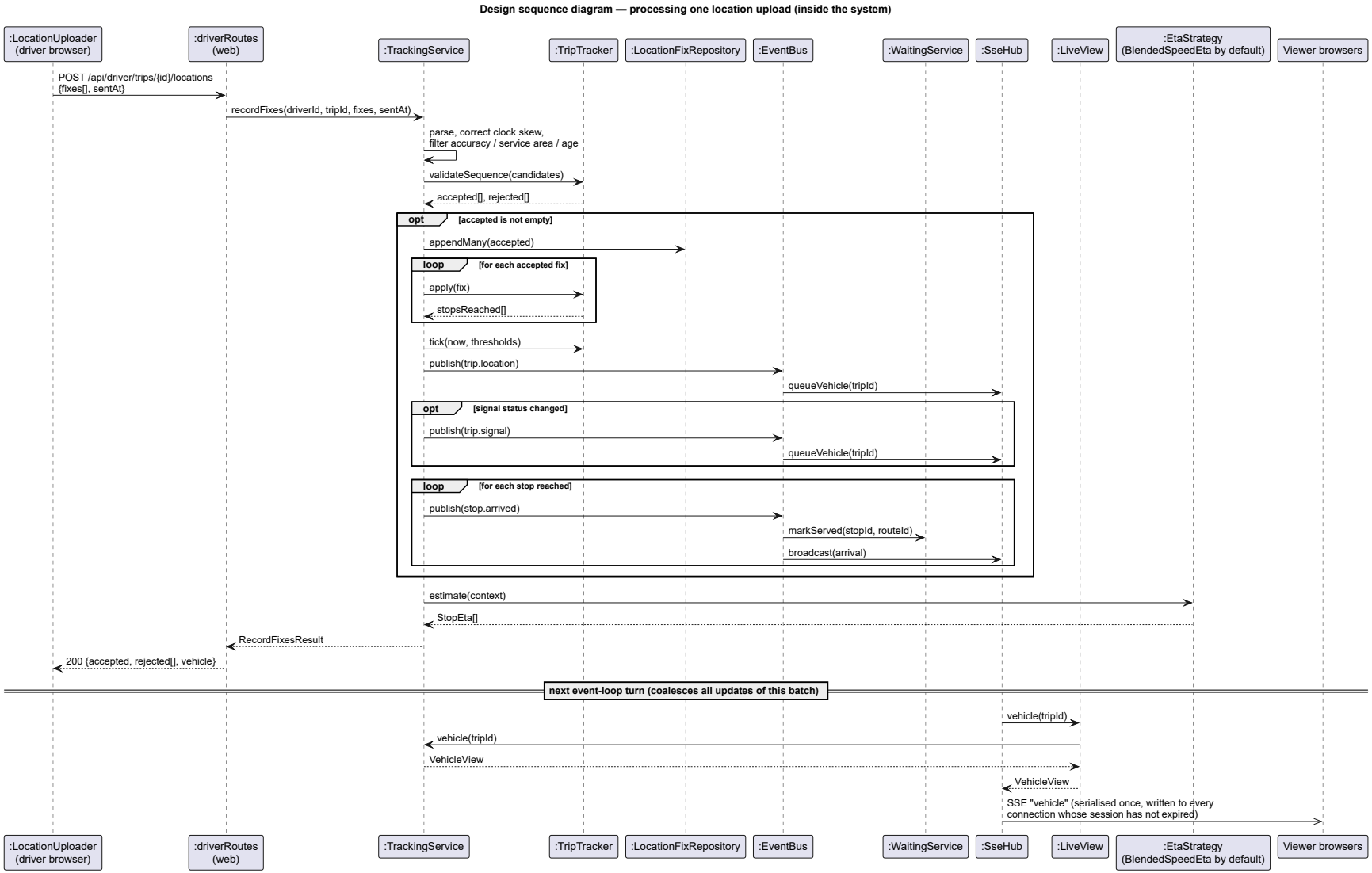


Figure A.5 — Inside the system: one location upload — overview at 49 % (read the 2 detail pages that follow, at 67 %; on screen, zoom in)

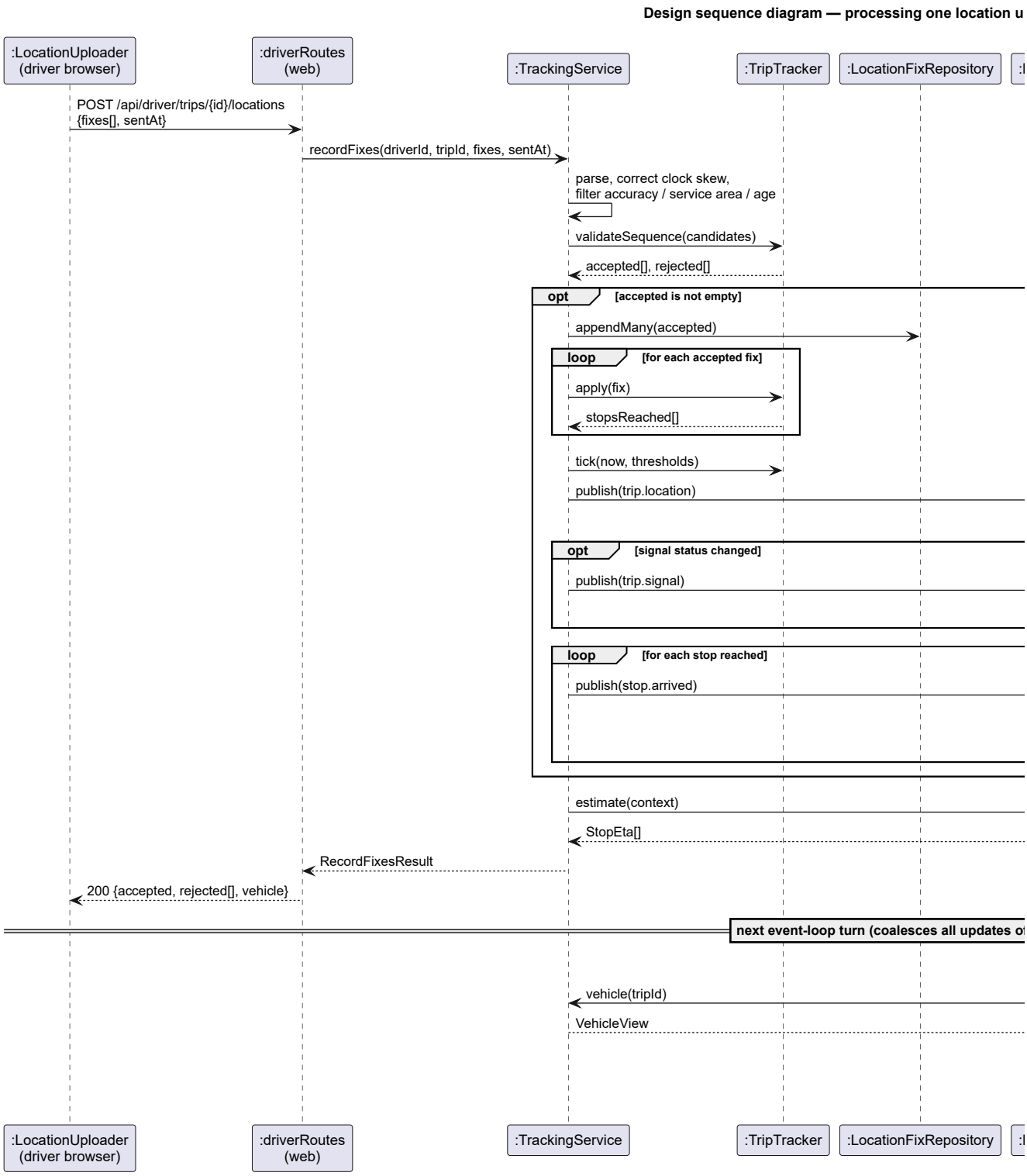


Figure A.5 — Inside the system: one location upload — detail 1 of 2 (column 1 of 2), at 67 %

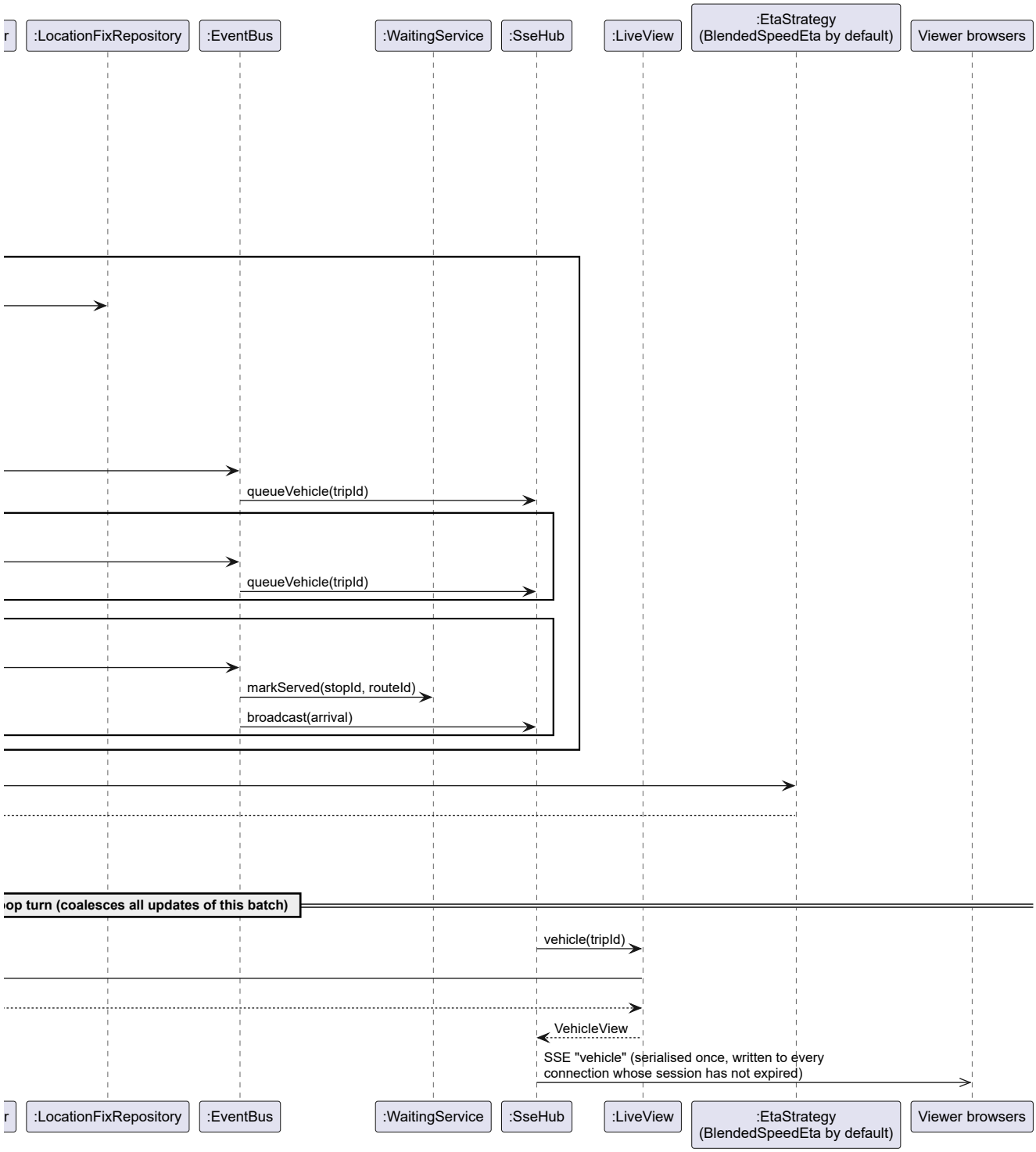


Figure A.5 — Inside the system: one location upload — detail 2 of 2 (column 2 of 2), at 67 %

A.6 State machine — Trip and GPS signal

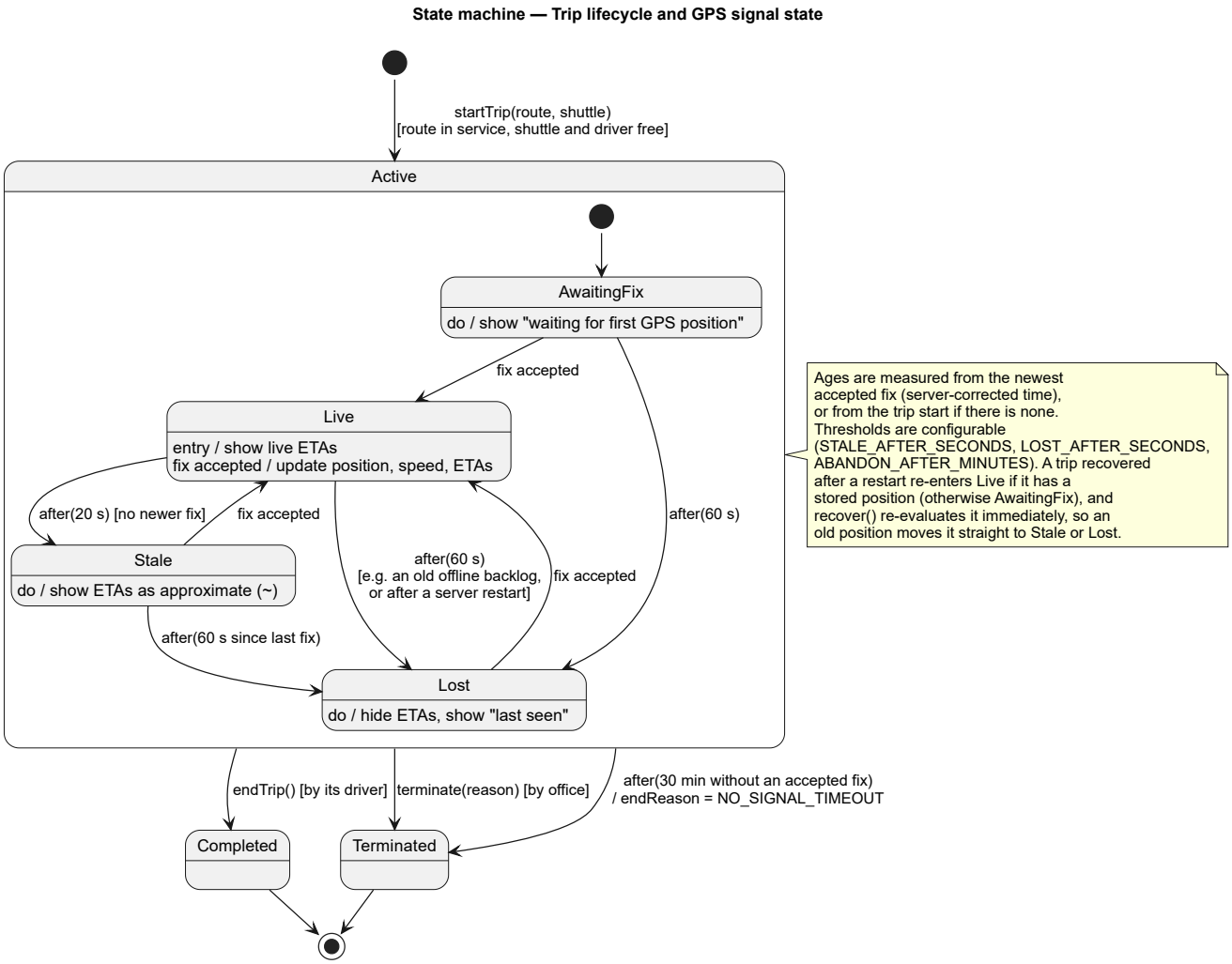


Figure A.6 — State machine for a trip and its GPS signal

`Active` is a composite state; its sub-states are implemented one-to-one by the State pattern classes in `src/tracking/signal-state.ts` (Figure A.15). Time events use UML `after(...)` notation.

A.7 State machine — Waiting request

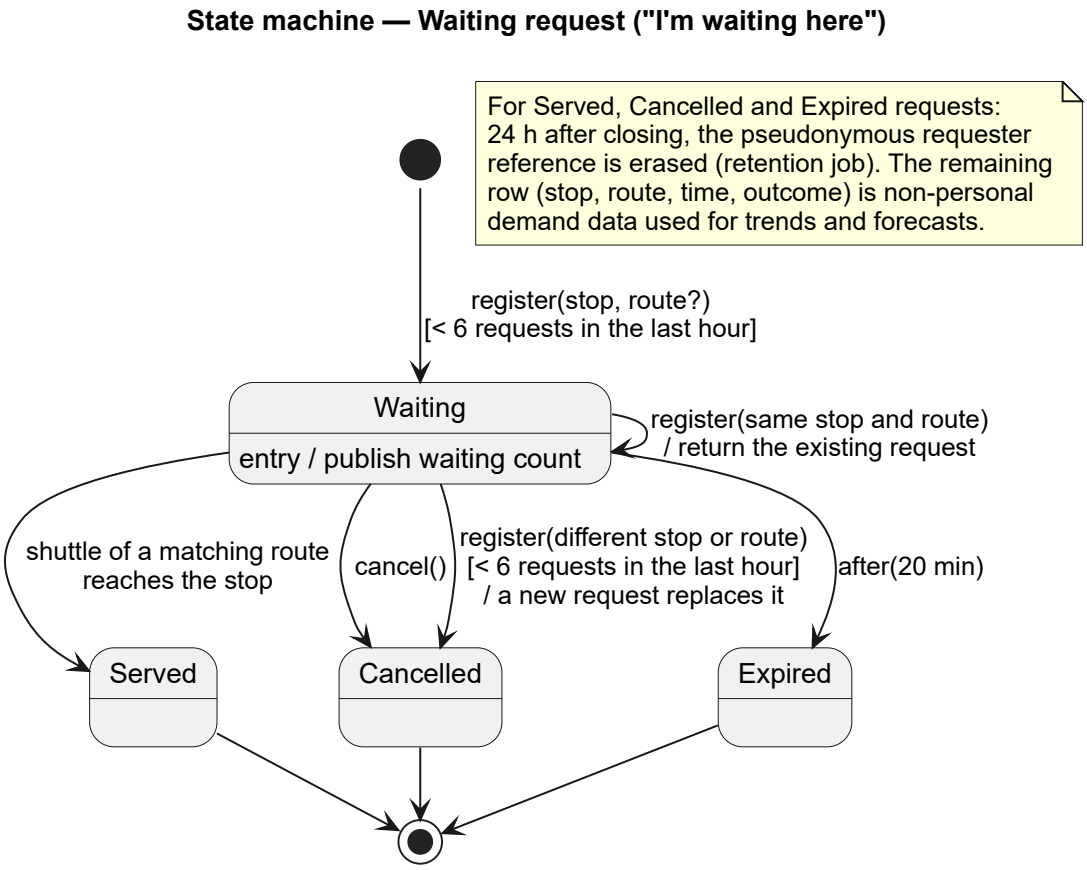


Figure A.7 — State machine for a waiting request

A.8 Domain model

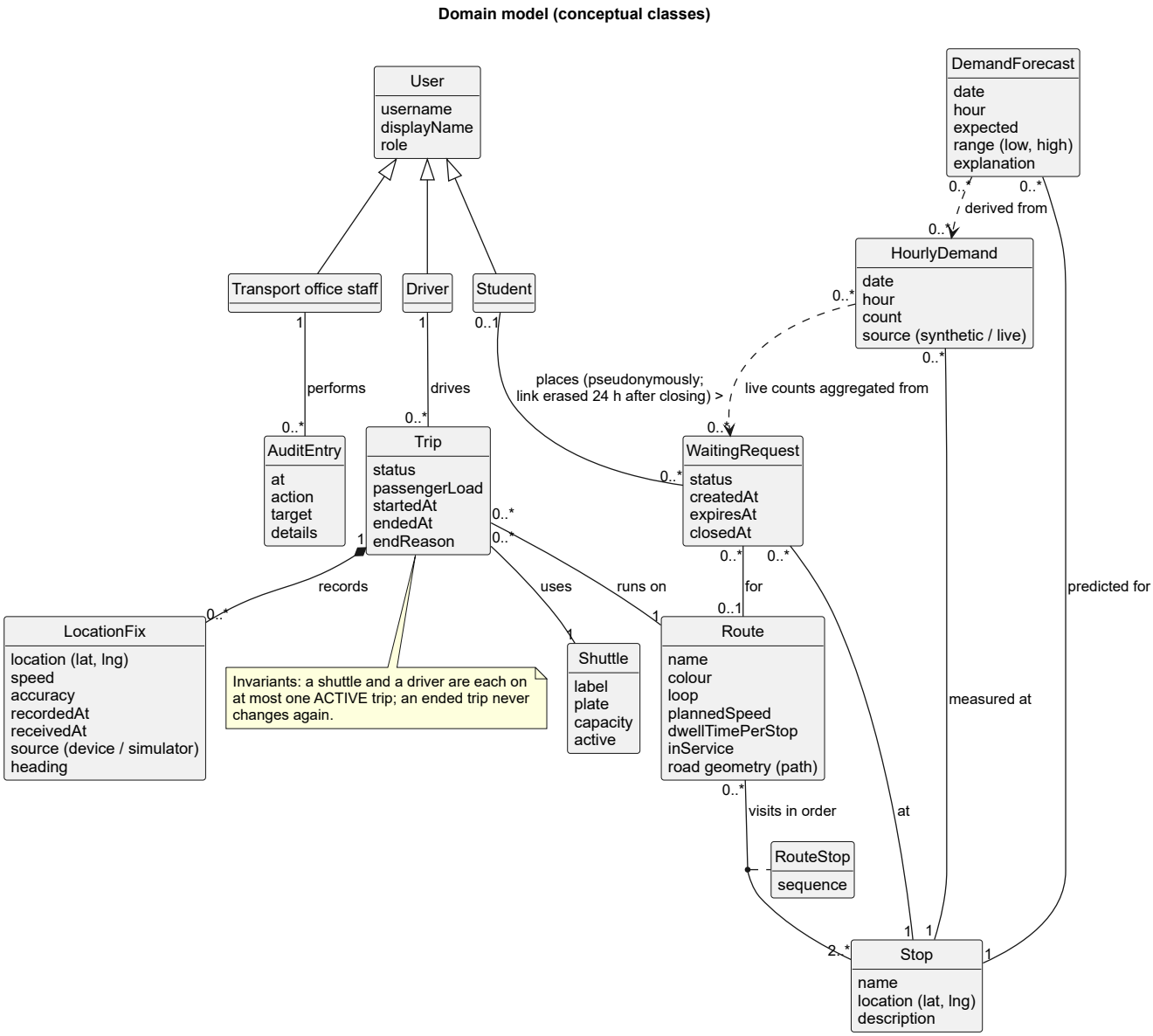


Figure A.8 — Domain model

Conceptual classes only (no operations). `RouteStop` is an association class carrying the stop order of a route.

A.9 Design class diagram 1 of 2 — tracking, network, real-time

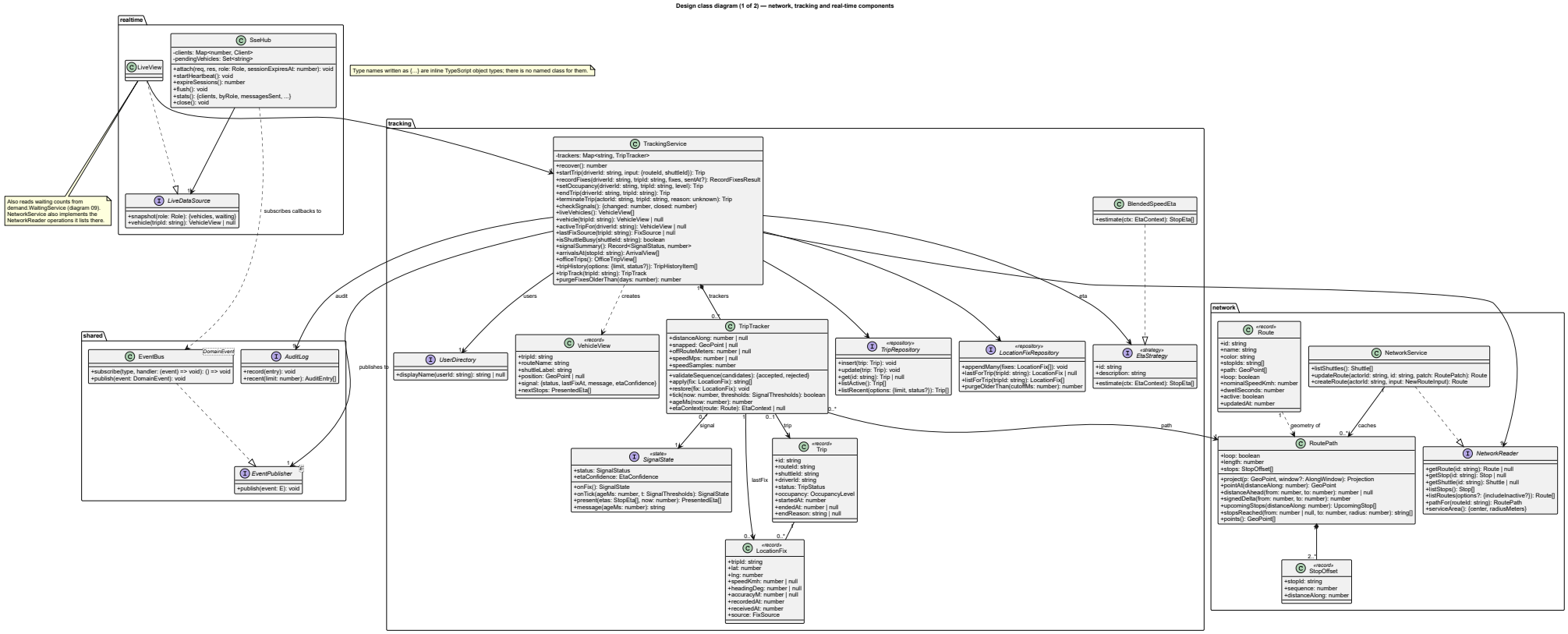


Figure A.9 — Design class diagram: tracking, network and real-time — overview at 28 % (read the 4 detail pages that follow, at 63 %; on screen, zoom in)

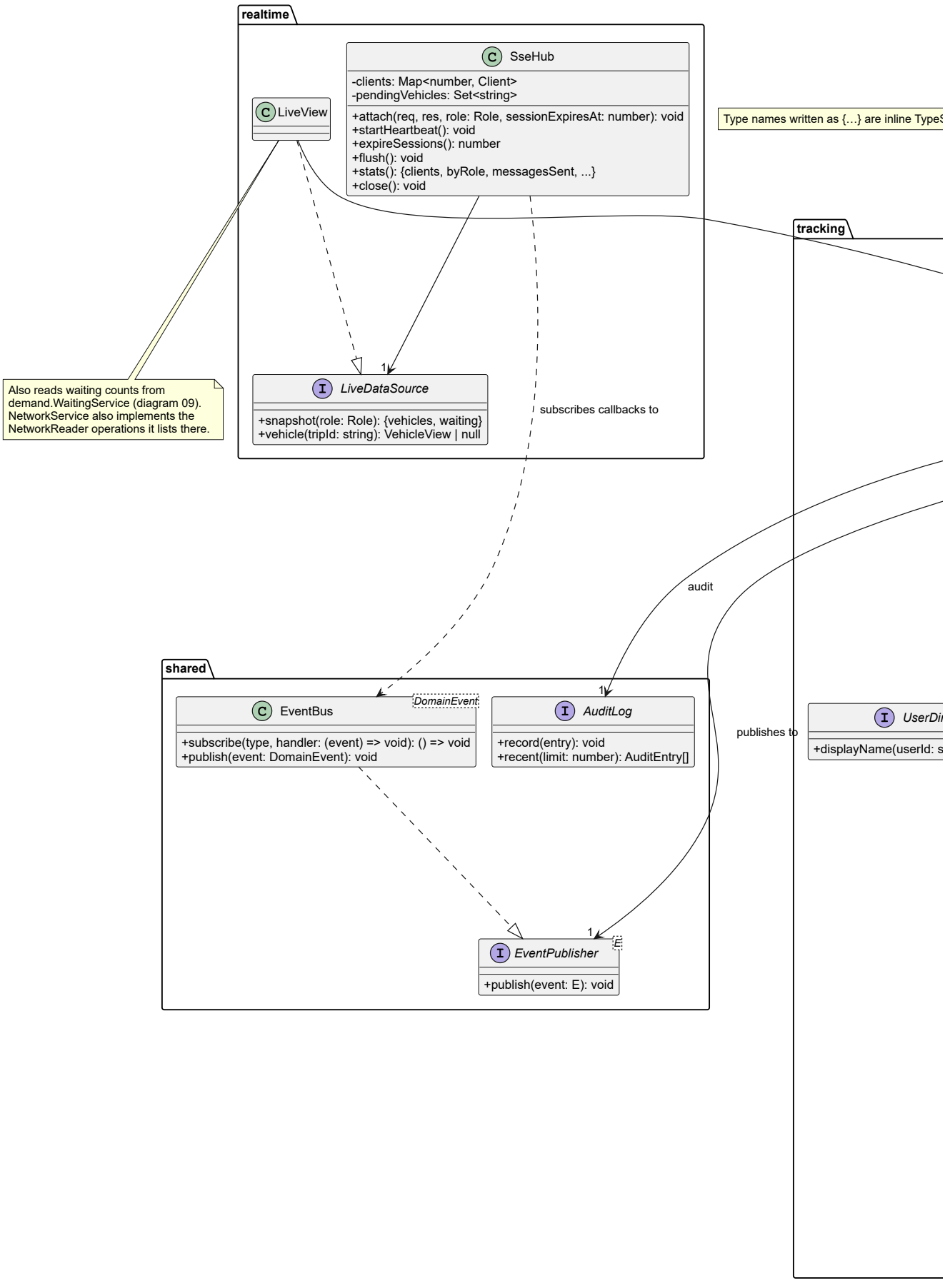


Figure A.9 — Design class diagram: tracking, network and real-time — detail 1 of 4 (column 1 of 4), at 63 %

mes written as {…} are inline TypeScript object types; there is no named class for them.

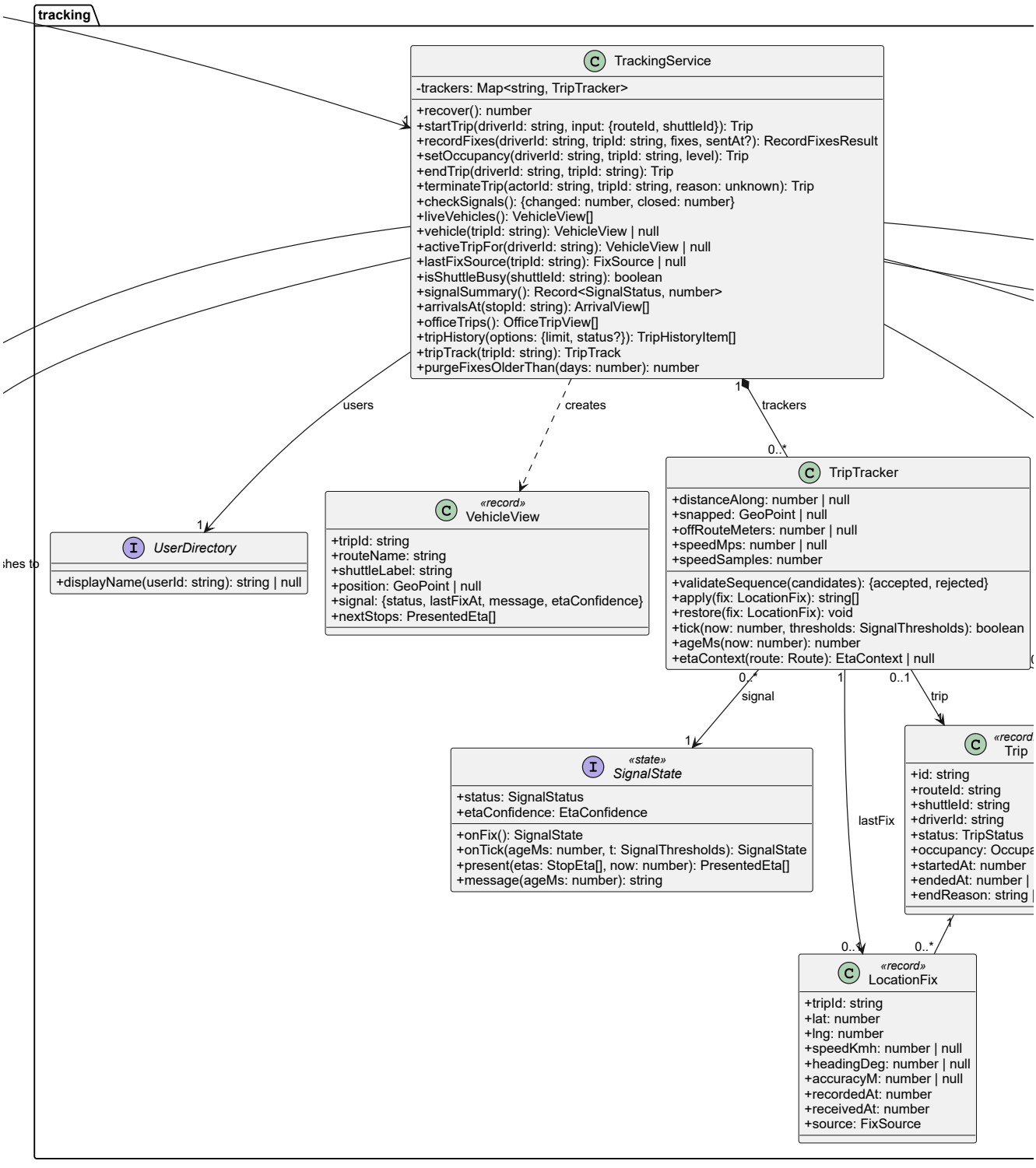


Figure A.9 — Design class diagram: tracking, network and real-time — detail 2 of 4 (column 2 of 4), at 63 %

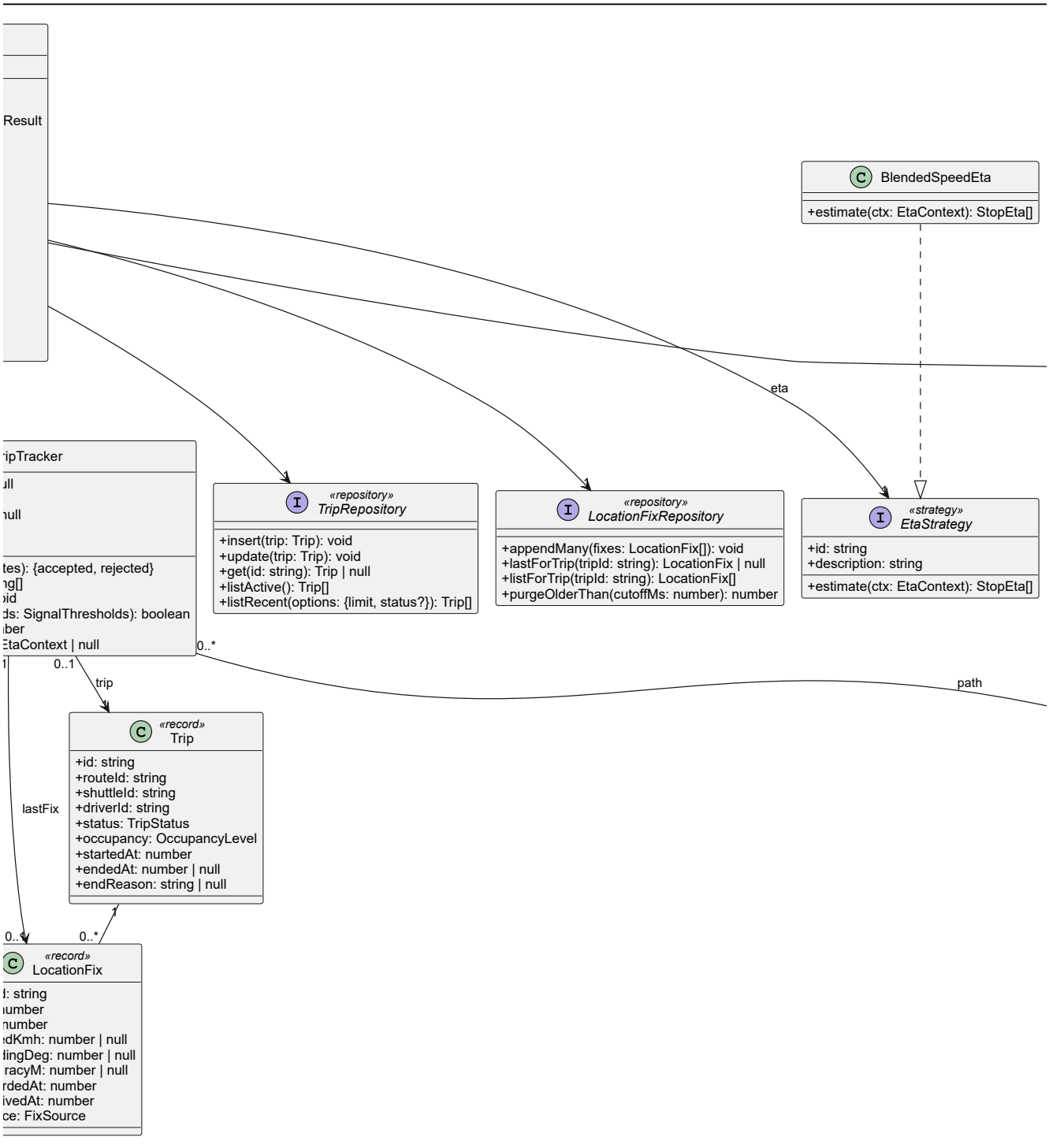


Figure A.9 — Design class diagram: tracking, network and real-time — detail 3 of 4 (column 3 of 4), at 63 %

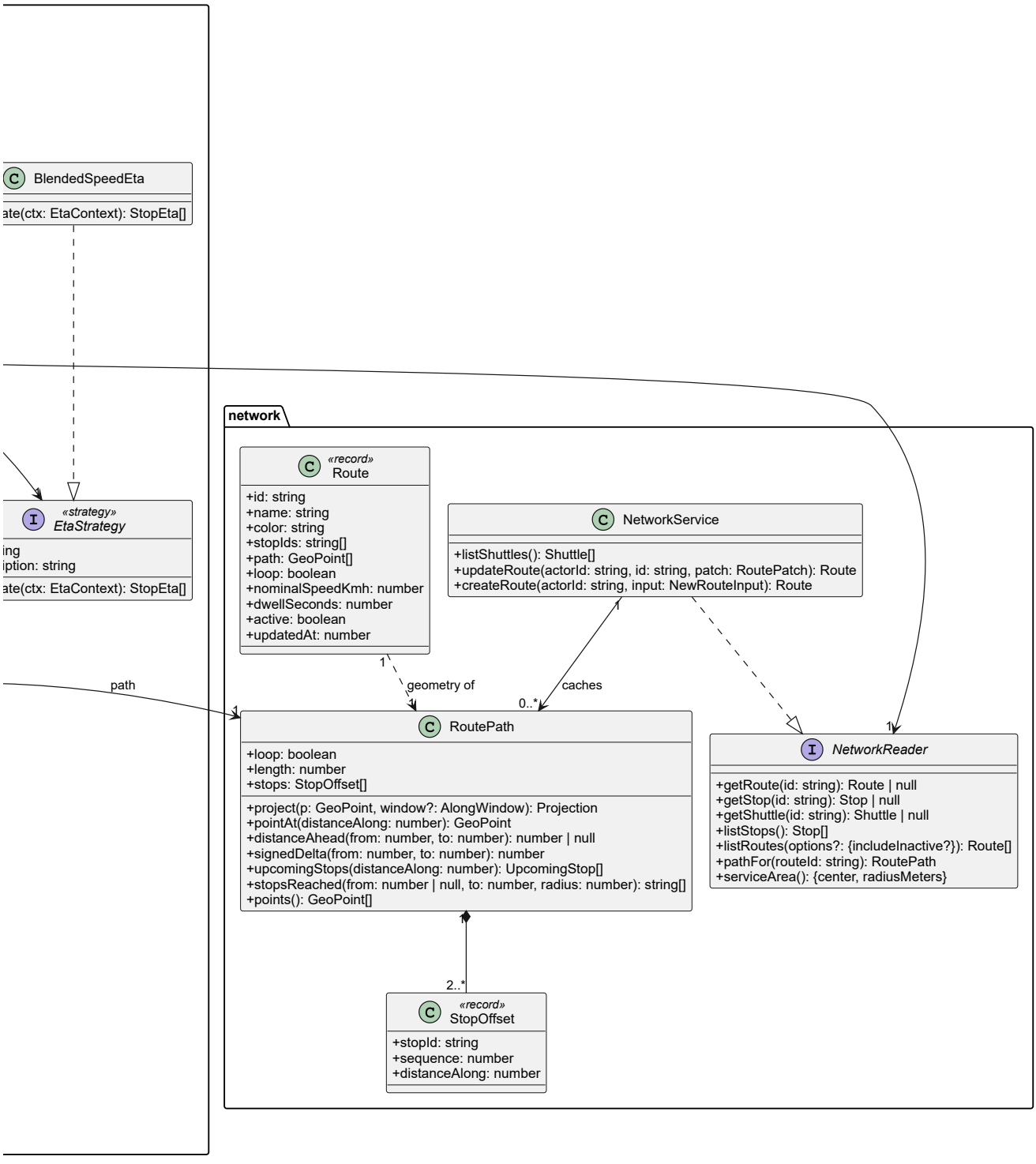


Figure A.9 — Design class diagram: tracking, network and real-time — detail 4 of 4 (column 4 of 4), at 63 %

A.10 Design class diagram 2 of 2 — demand, forecasting, identity

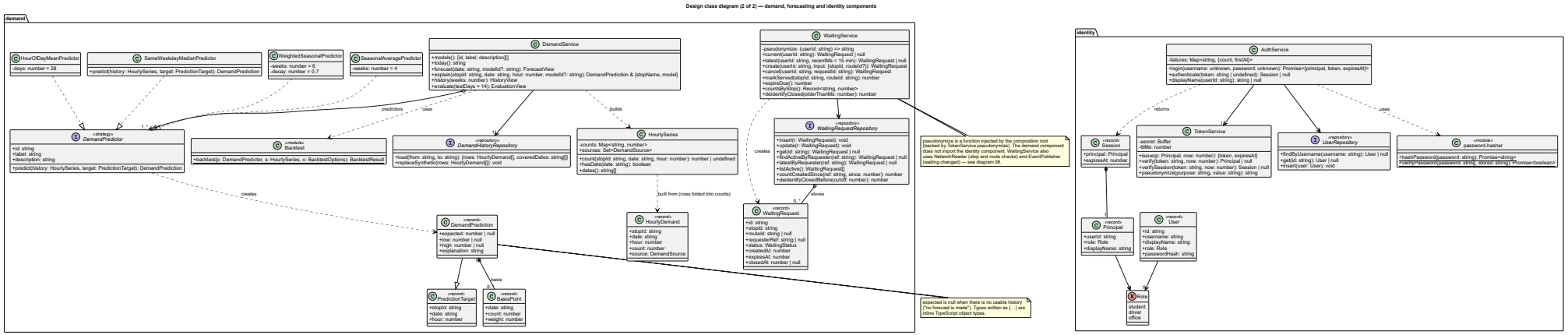


Figure A.10 — Design class diagram: demand, forecasting and identity — overview at 24 % (read the 3 detail pages that follow, at 65 %; on screen, zoom in)

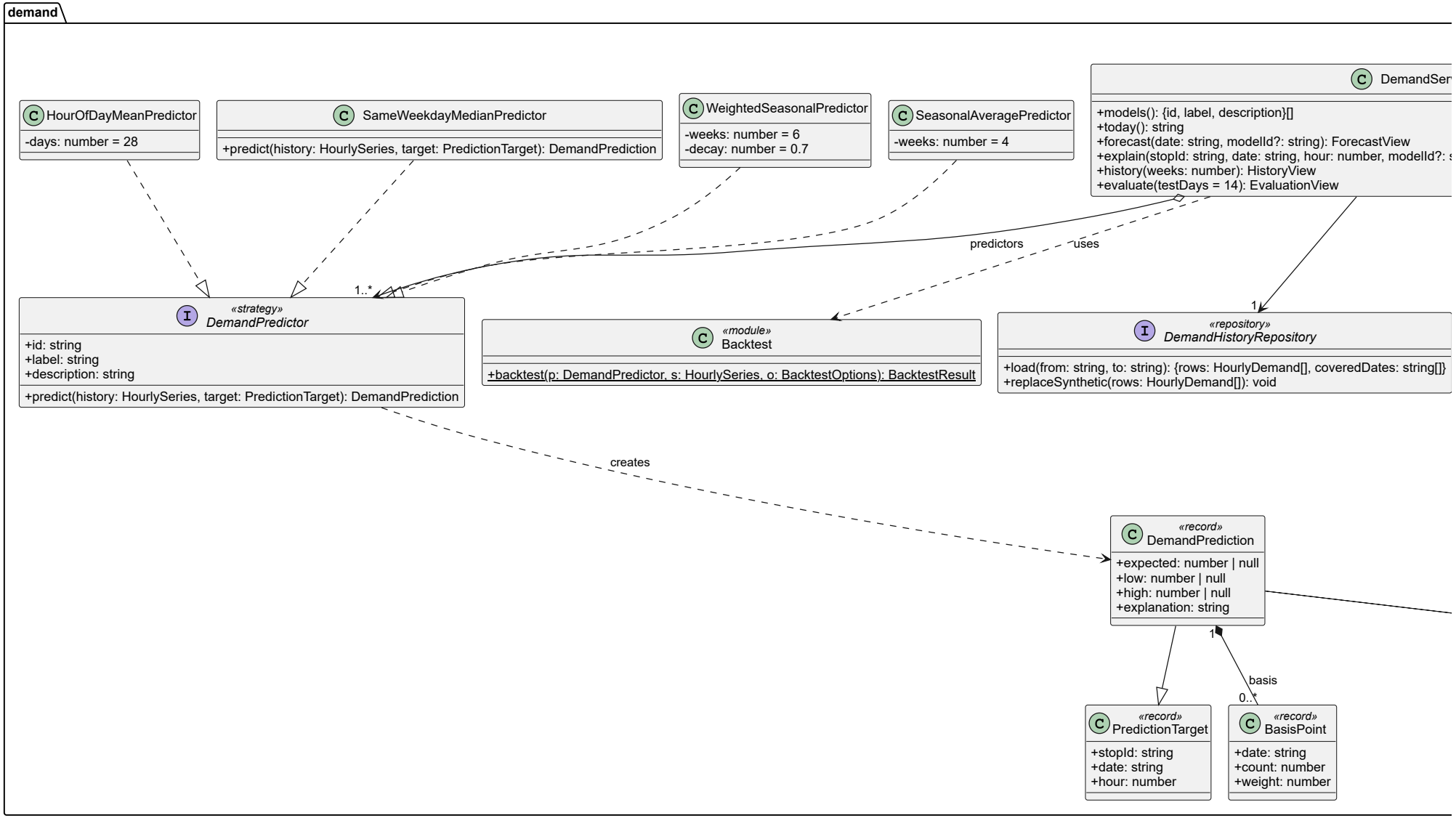


Figure A.10 — Design class diagram: demand, forecasting and identity — detail 1 of 3 (column 1 of 3), at 65 %

Design class diagram (2 of 2) — demand, forecasting and identity components

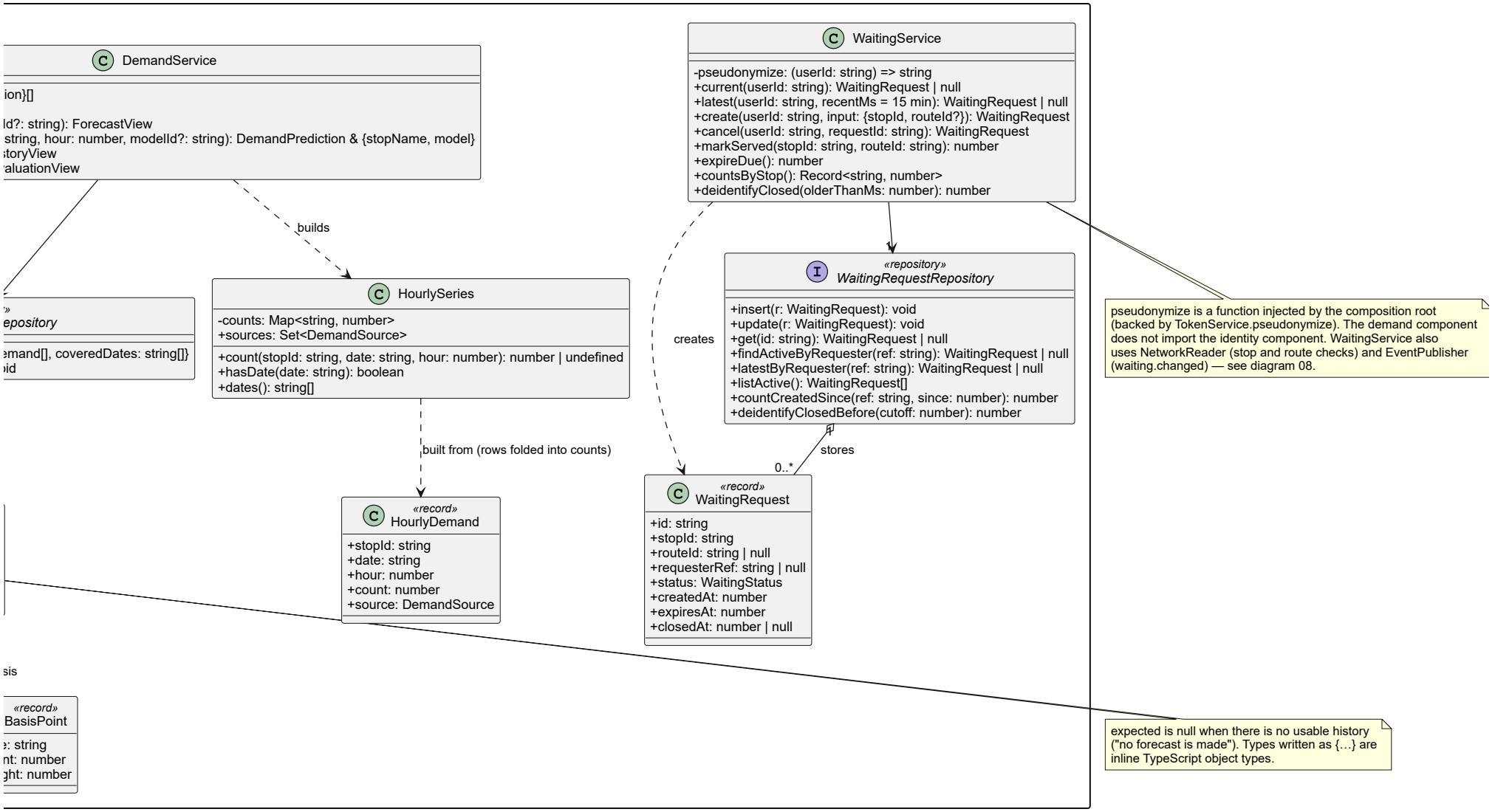


Figure A.10 — Design class diagram: demand, forecasting and identity — detail 2 of 3 (column 2 of 3), at 65 %

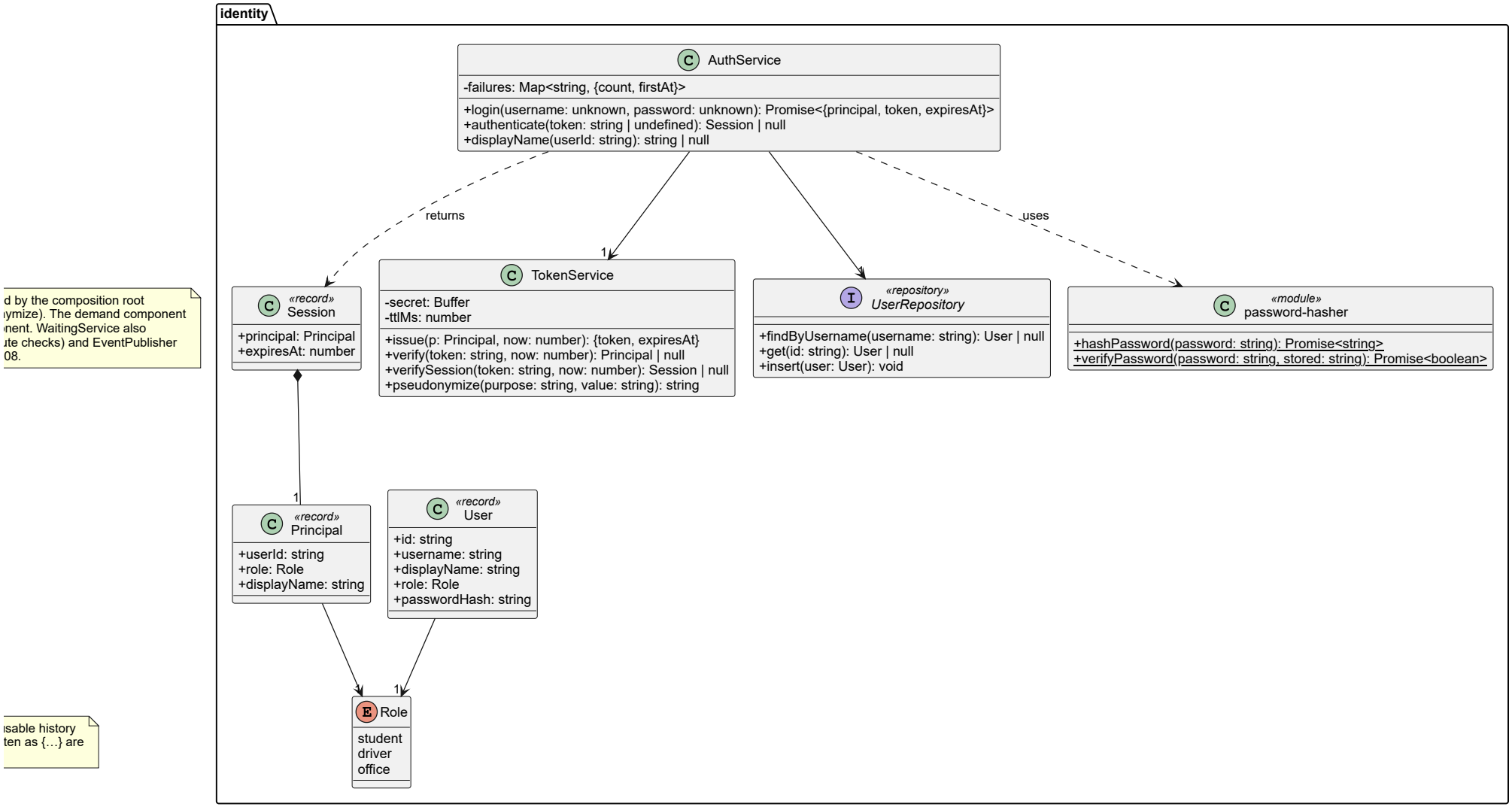


Figure A.10 — Design class diagram: demand, forecasting and identity — detail 3 of 3 (column 3 of 3), at 65 %

A.11 Component diagram

Each component is a folder under `src/` with a public service class and port interfaces. Provided interfaces are drawn as lollipops, required ones as dependencies on them.

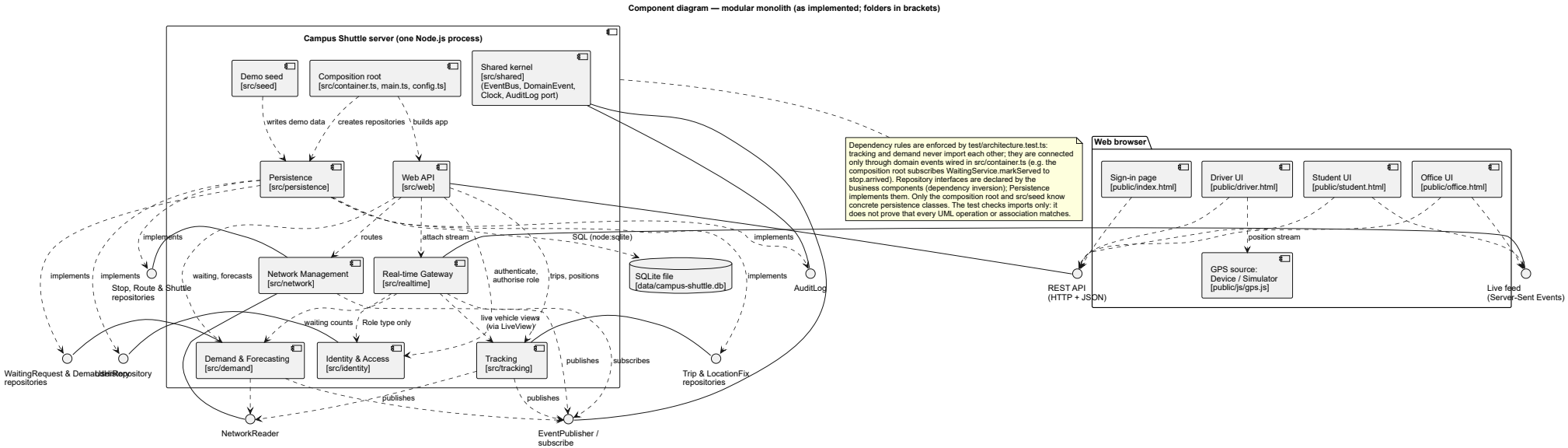


Figure A.11 — Component diagram — overview at 38 % (read the 2 detail pages that follow, at 71 %; on screen, zoom in)

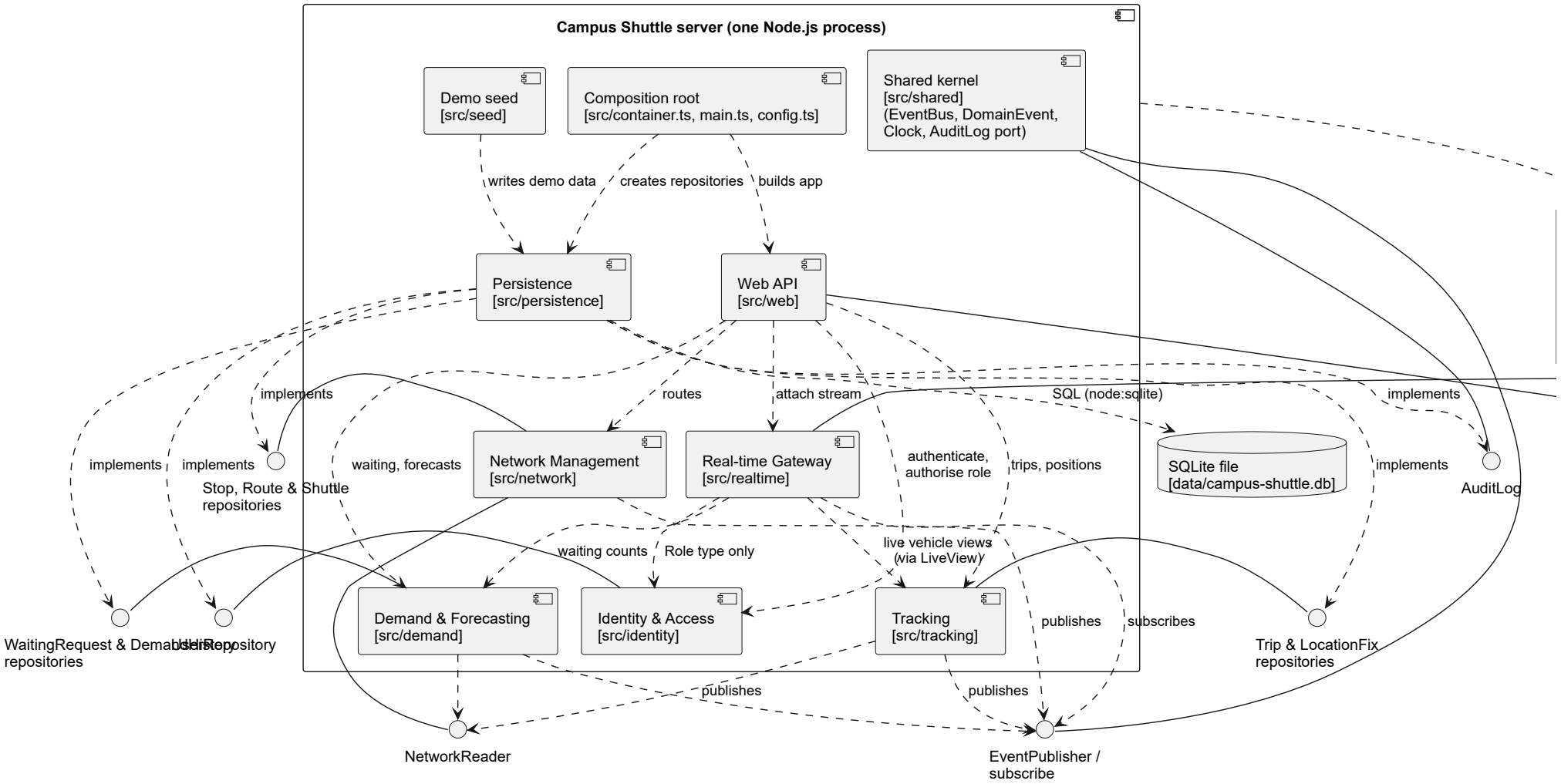


Figure A.11 — Component diagram — detail 1 of 2 (column 1 of 2), at 71 %

ıodular monolith (as implemented; folders in brackets)

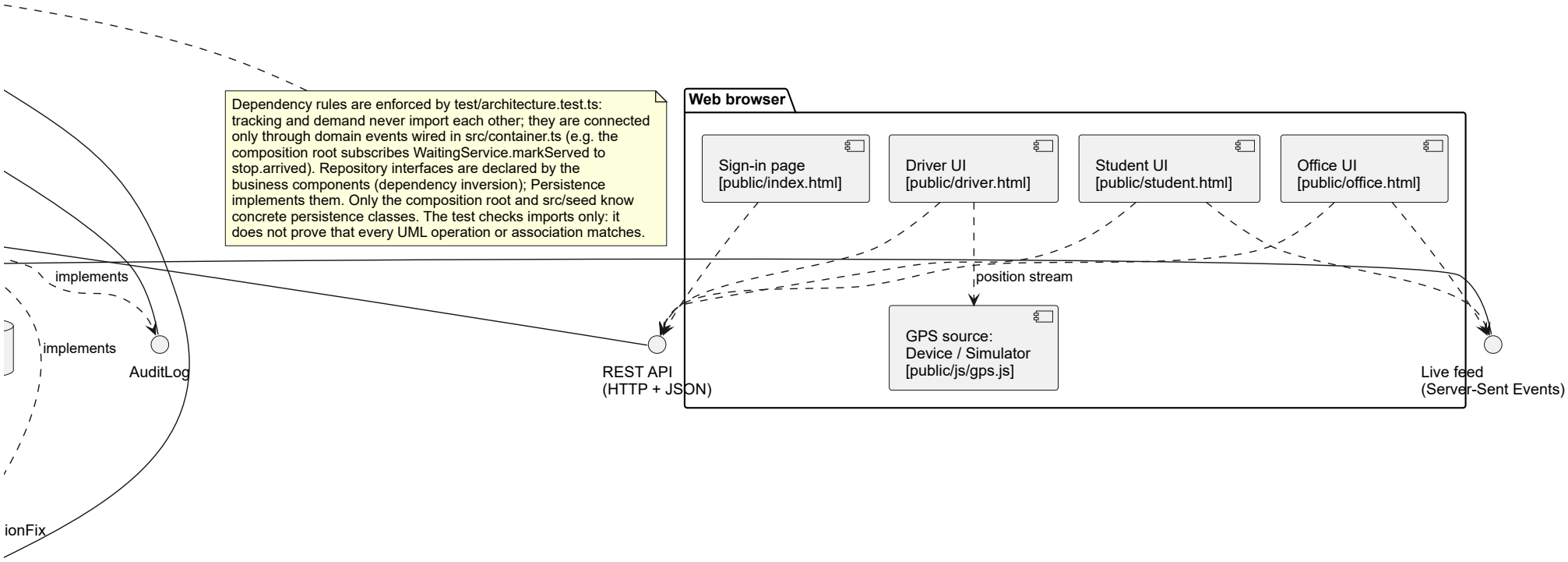


Figure A.11 — Component diagram — detail 2 of 2 (column 2 of 2), at 71 %

A.12 Deployment diagram

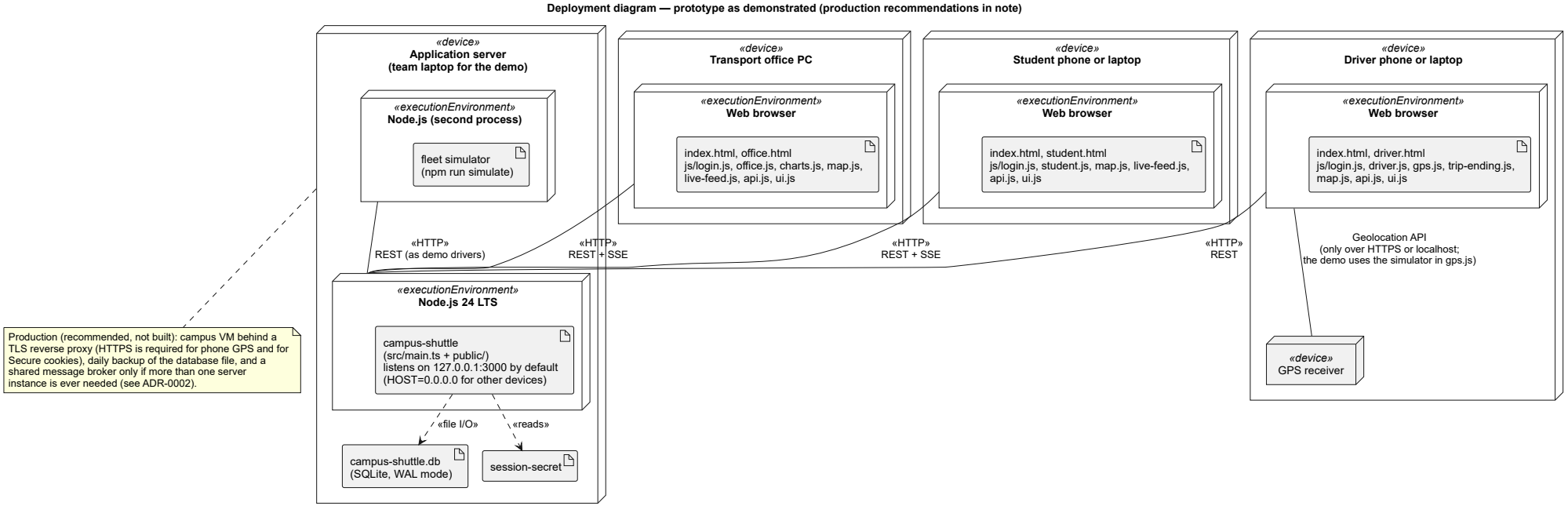


Figure A.12 — Deployment diagram — overview at 51 % (read the 2 detail pages that follow, at 93 %; on screen, zoom in)

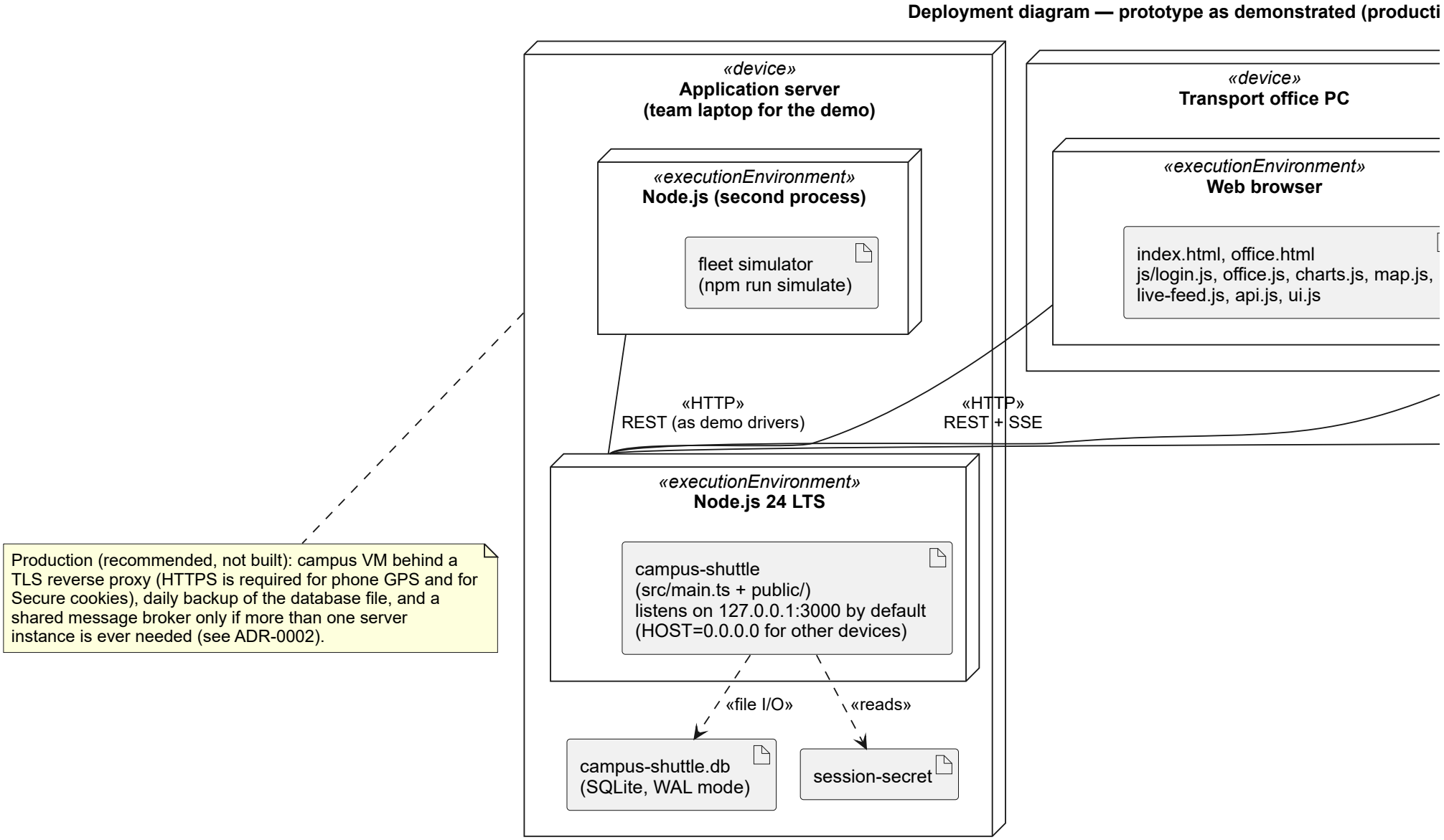


Figure A.12 — Deployment diagram — detail 1 of 2 (column 1 of 2), at 93 %

ype as demonstrated (production recommendations in note)

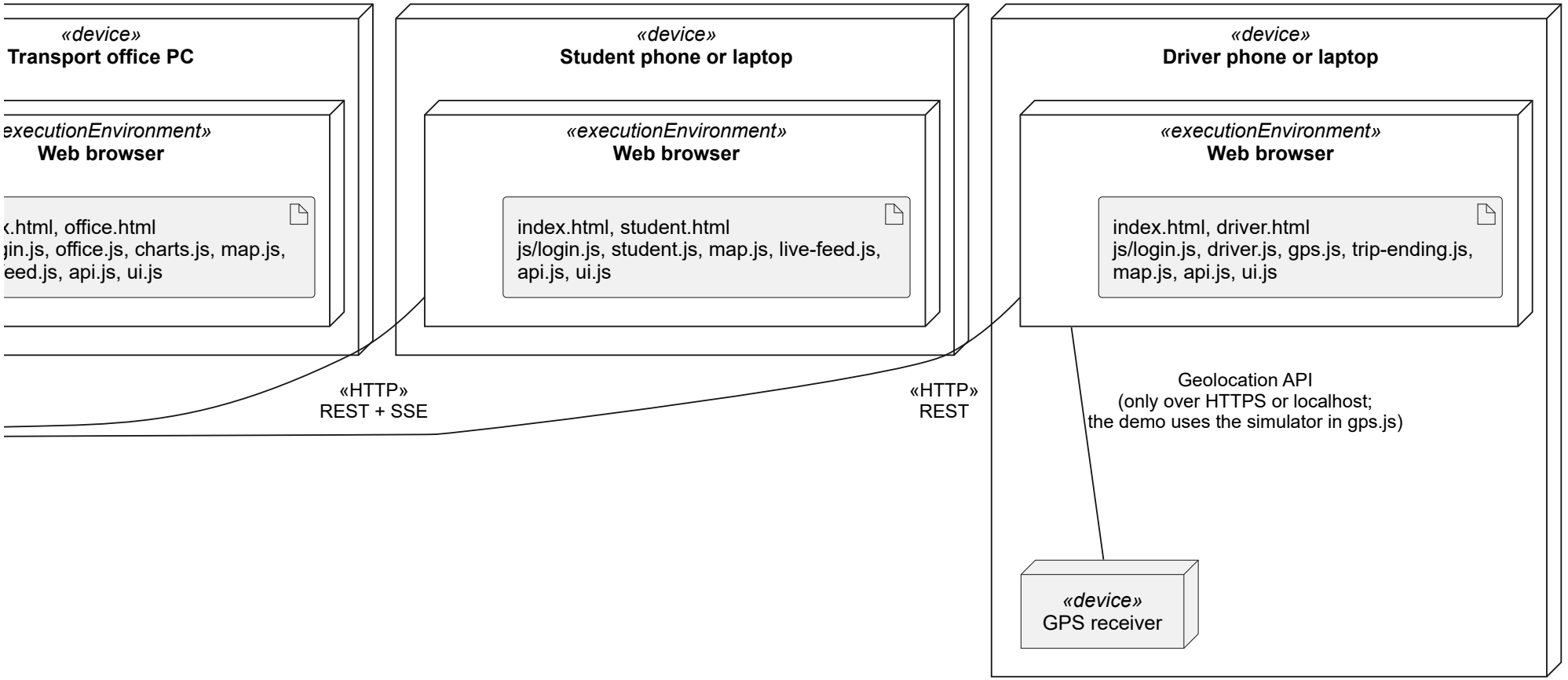


Figure A.12 — Deployment diagram — detail 2 of 2 (column 2 of 2), at 93 %

A.13–A.17 Design pattern class diagrams

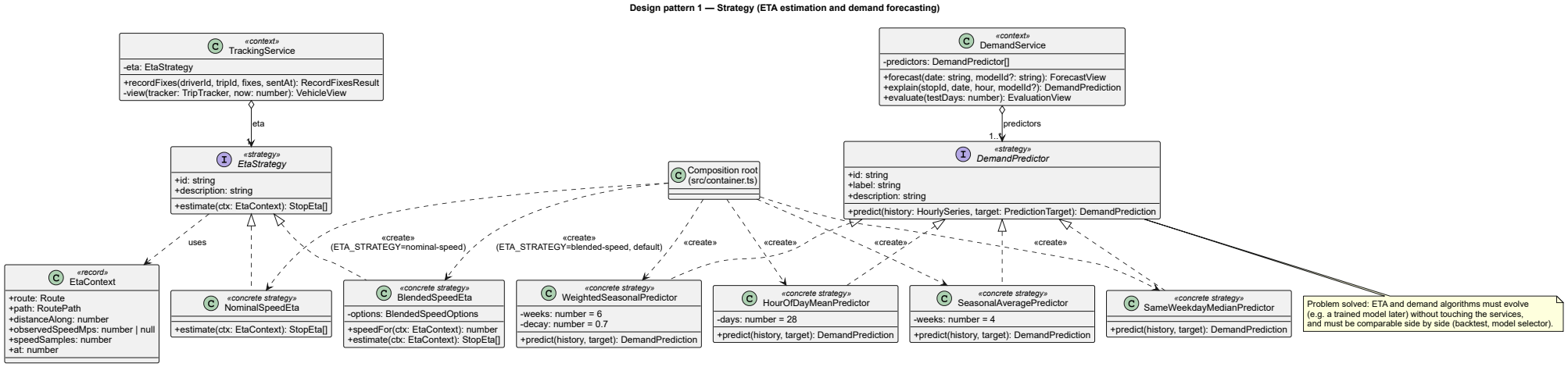


Figure A.13 — Strategy pattern — overview at 43 % (read the 2 detail pages that follow, at 79 %; on screen, zoom in)

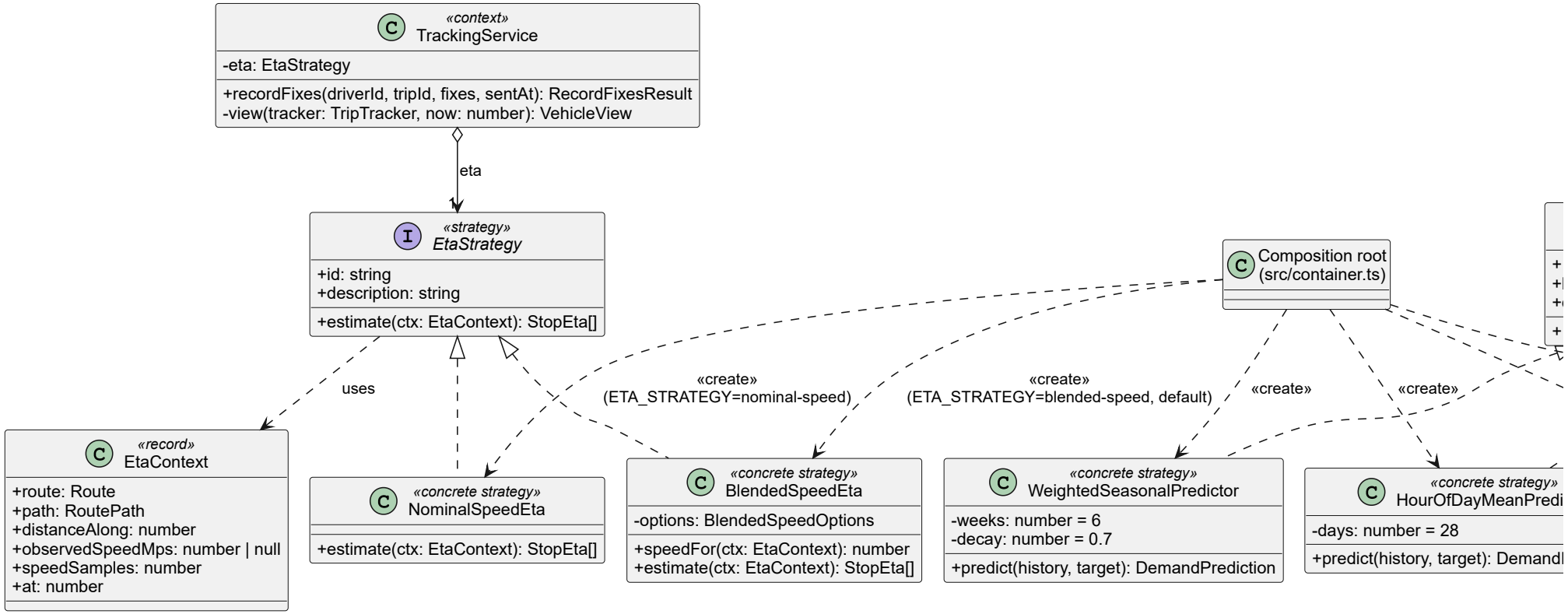


Figure A.13 — Strategy pattern — detail 1 of 2 (column 1 of 2), at 79 %

Strategy (ETA estimation and demand forecasting)

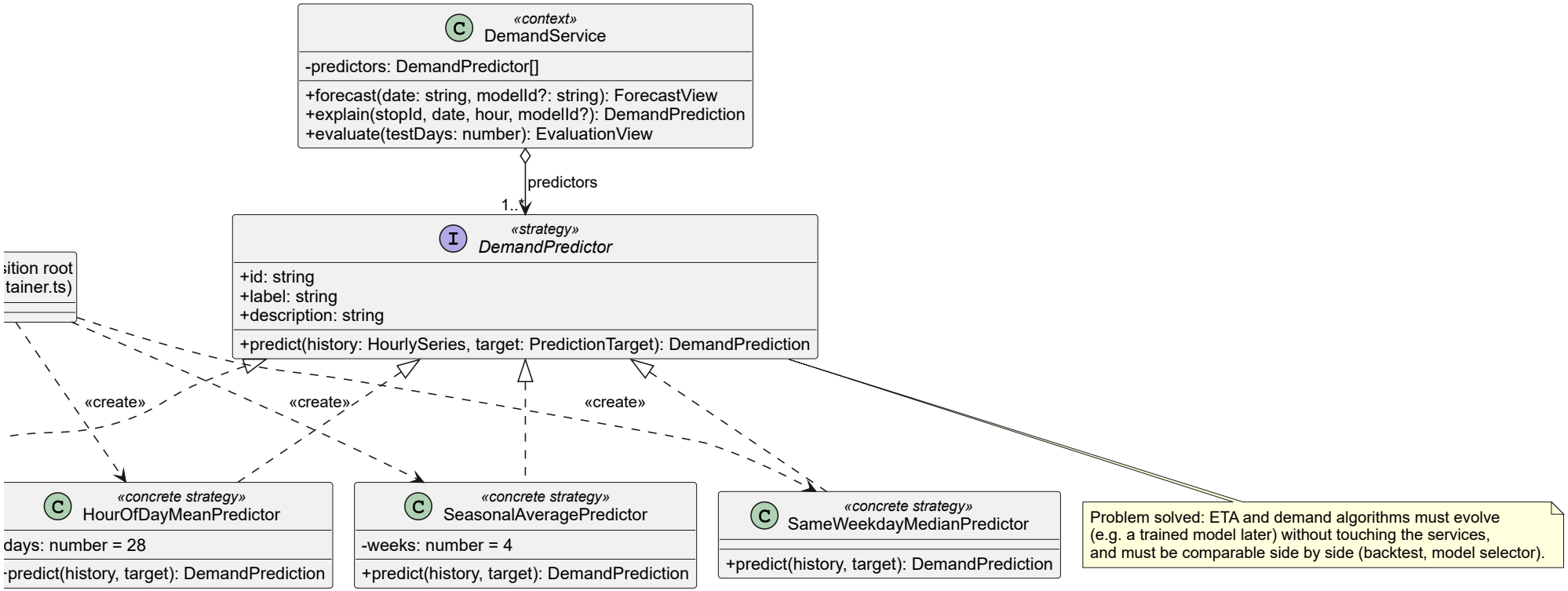


Figure A.13 — Strategy pattern — detail 2 of 2 (column 2 of 2), at 79 %

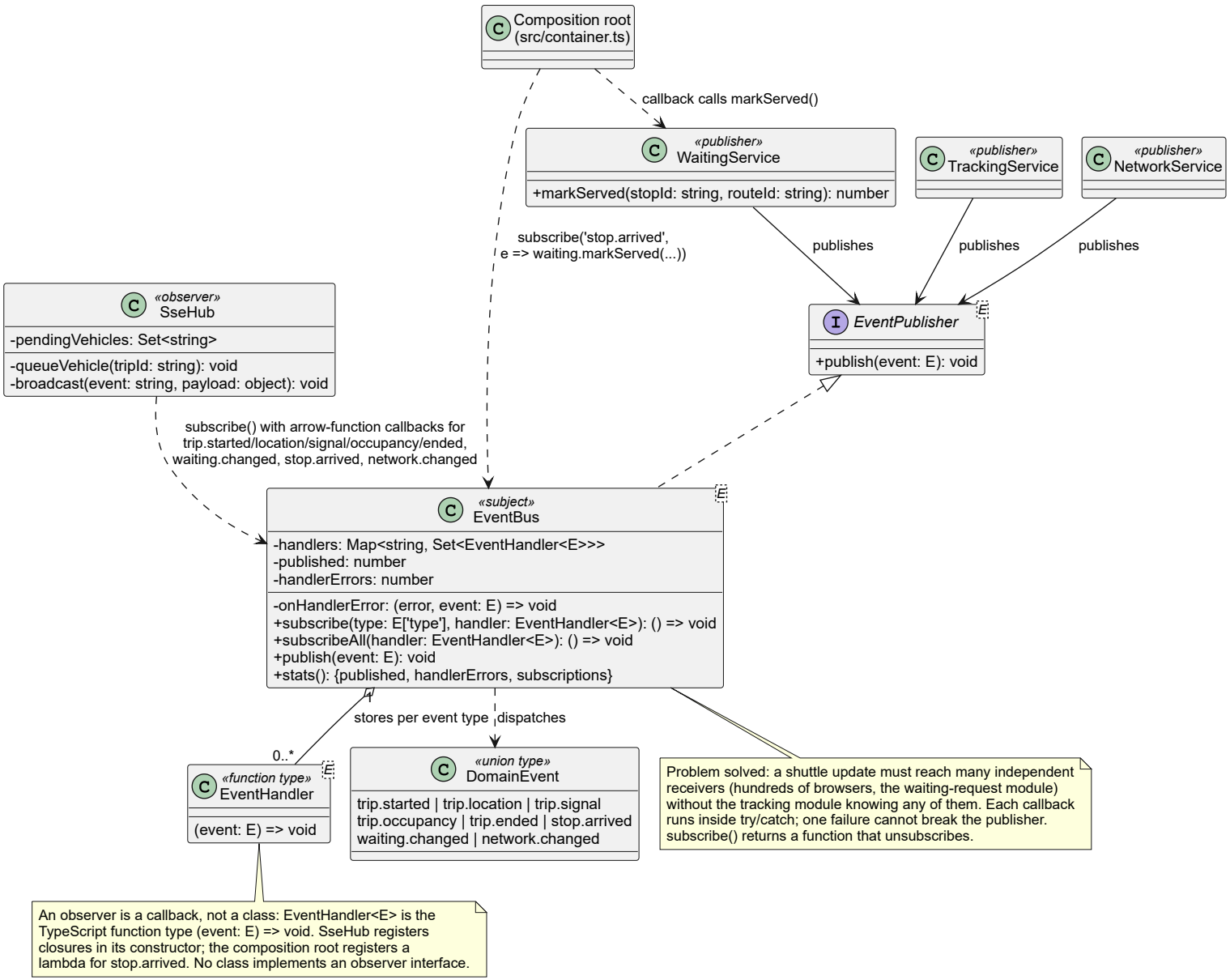


Figure A.14 — Observer pattern

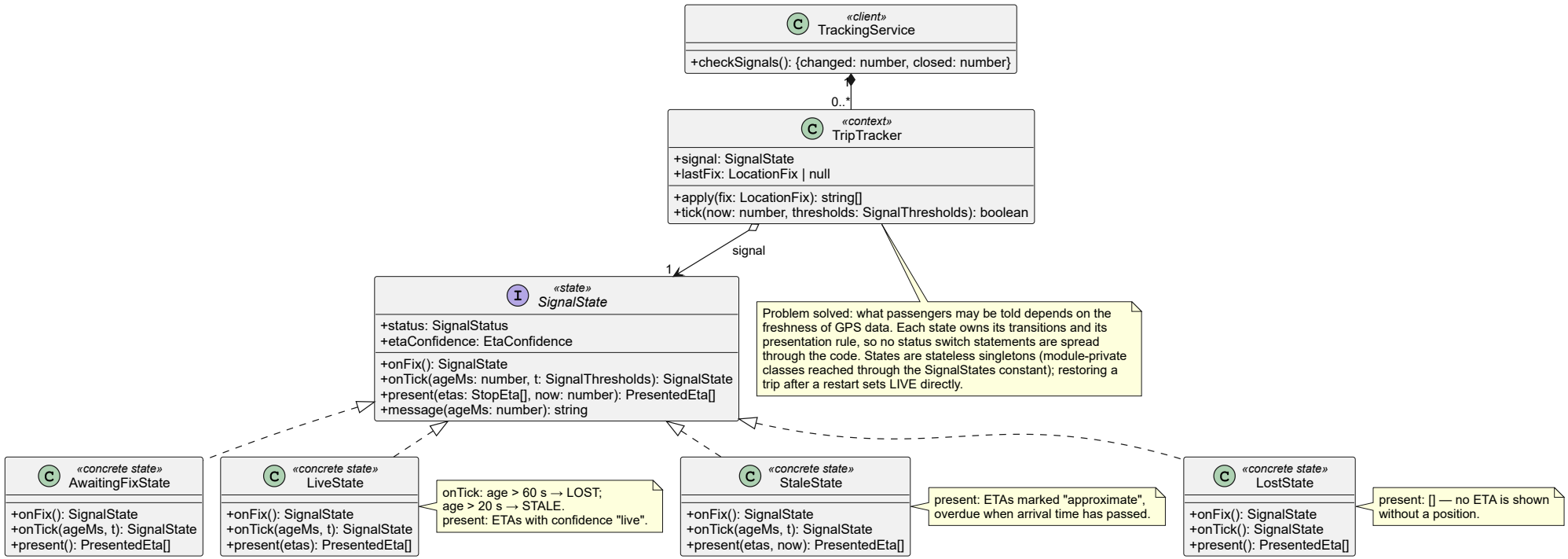


Figure A.15 — State pattern

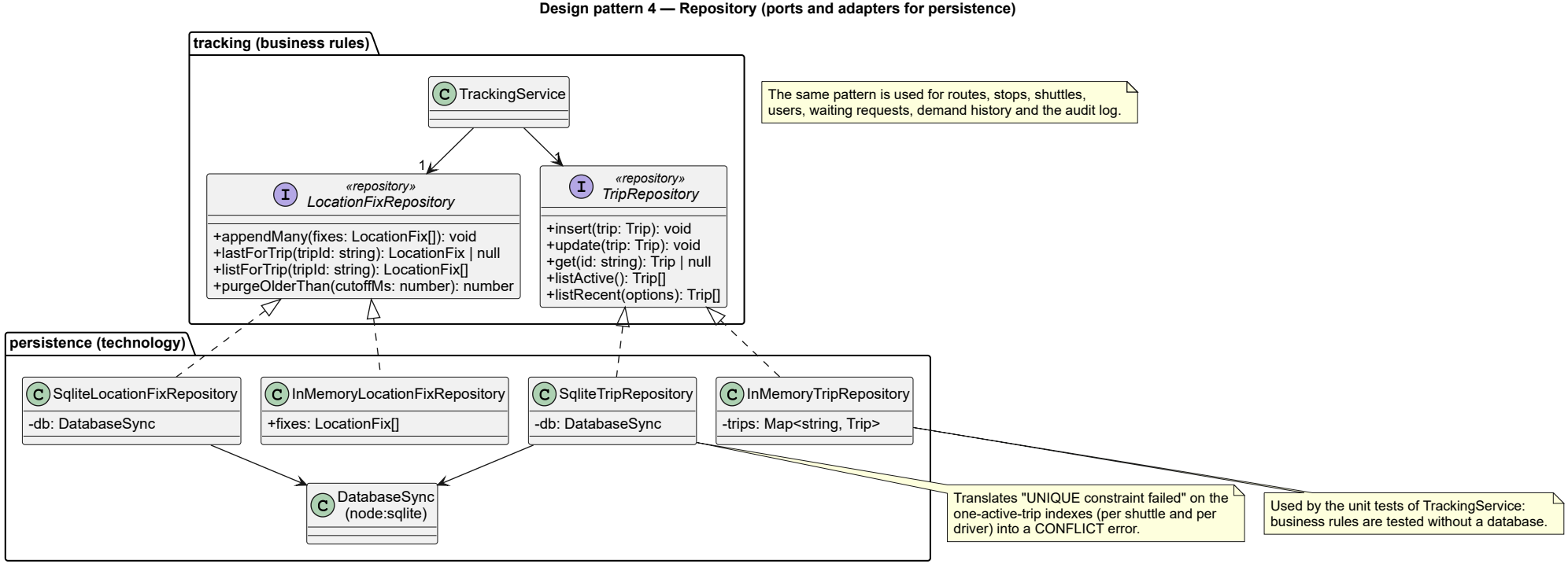


Figure A.16 — Repository pattern

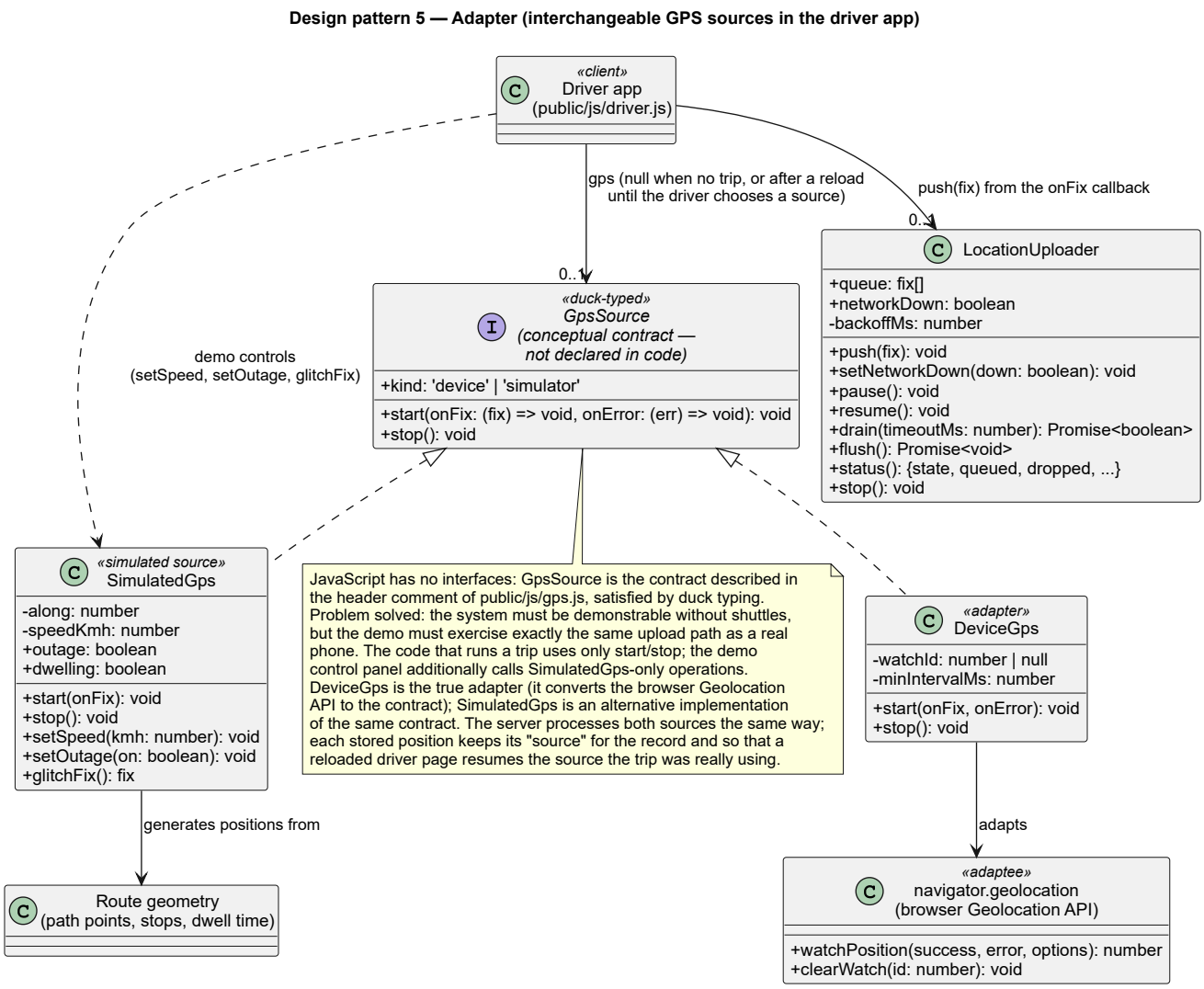


Figure A.17 — Adapter pattern